Draft version June 7, 2026
Typeset using LATEX twocolumn style in AASTeX631

A Probabilistic Gaia DR3 Catalogue of Stars with Very Low Galactic Rest-Frame Speeds

Rodney Humble[1]

[1] *Independent Researcher, San Diego, CA, USA*

ABSTRACT

We present a Gaia DR3 catalogue of stars selected by very low Galactic rest-frame speed, $V_{\rm grf} < 25\,{\rm km\,s^{-1}}$. Using Lindegren et al. (2021) parallax zero-point corrections, Bailer-Jones et al. (2021) photogeometric distances, and Monte Carlo propagation of the Gaia parallax–proper-motion covariance, we identify a primary Tier A+B sample of 541 stars with $P(V_{\rm grf} < 25\,{\rm km\,s^{-1}}) > 0.84$ and a broader Tier A+B+C sample of 1,952 stars with $P > 0.50$. Summing $1 - P$ over the members, the internal Monte Carlo scores imply a nominal expected membership purity of $94\% \pm 1\%$ for Tier A+B and $73\% \pm 1\%$ for Tier A+B+C. Orbit integrations in the static, axisymmetrized Hunter et al. (2024) Galactic potential map the broader sample onto compact, highly radial model orbits, with median eccentricity 0.949, median pericentre 154 pc, and median apocentre 7.35 kpc. In an observed-control extrapolation, the Tier A+B+C count exceeds the tested control-normalised smooth velocity nulls by a factor of several (7.4× for the conservative Gaussian-mixture null; 25.6× for an illustrative single ellipsoid), and unweighted Anderson–Darling tests reject continuity with the adjacent 25–50 $\rm km\,s^{-1}$ band ($p < 0.001$). A controlled GeDR3mock injection-recovery exercise recovers 60.9% of true sub-$25\,{\rm km\,s^{-1}}$ stars end-to-end (87.6% conditional on entering the pipeline). The dominant selection-function systematic affects deprojection rather than catalogue membership: 83.2% of valid Tier A+B sources fall in low-parent-count ($n < 10$) GaiaUnlimited cells, and including prior-filled cells changes the Tier A+B effective-volume scale by a factor of 2.09. We release the catalogue as a probabilistic kinematic sample with model-dependent orbit diagnostics and subset-qualified chemical context.

Keywords: Milky Way Galaxy (1054) — Milky Way dynamics (1051) — Galaxy structure (622) — Stellar kinematics (1608) — Astrometry (80) — Sky surveys (1464) — Catalogs (205)

1. INTRODUCTION

Gaia DR3 provides full six-dimensional phase-space information for tens of millions of stars through its astrometry and radial velocities (Gaia Collaboration et al. 2023; Katz et al. 2023). This makes it possible to search the observed Milky Way phase-space distribution for rare kinematic states, not only for previously defined stellar populations or merger debris. Here we focus on one such state: stars observed with unusually low instantaneous Galactic rest-frame speed. The broader Galactic-structure and Gaia-era archaeology context is reviewed by Binney & Merrifield (1998), Bland-Hawthorn & Gerhard (2016), Helmi (2020), and Deason & Belokurov (2024). This complements Gaia-era catalogues selected at the opposite kinematic extreme, including Gaia DR3 hypervelocity-star searches (Marchetti et al. 2022; Carballo-Bello et al. 2025), and halo catalogues selected from high tangential velocity or reduced proper motion (Viswanathan et al. 2023). The closest published Gaia-era catalogue selected by low Galactocentric azimuthal velocity is Filion et al. (2025), which identifies $\sim$ 70 slow- and retrograde stars with thin-disc-like abundances. That work focuses on chemodynamic outliers within a disc-like chemistry box and is therefore complementary to the chemistry-agnostic, low-instantaneous-speed catalogue presented here.

The sample is defined by present-day kinematics, not by inferred origin. We construct a probabilistic low-$V_{\rm grf}$ Gaia DR3 catalogue, quantify the stability of the velocity-threshold membership under measurement uncertainties, and report model-dependent orbit diagnostics in an adopted Milky Way potential. The

rod.humble@gmail.com

orbit integrations are used to interpret the observed slow state, but they do not define the catalogue.

This framing is important because a very low instantaneous rest-frame speed is essentially a turning-point condition: the star must carry near-zero angular momentum and be caught close to an orbital turning point. More than one kind of orbit can satisfy it. The star may lie near the apocentre of a radial accretion orbit, as in the apocentre-pile-up interpretation of radial halo debris (Deason et al. 2018); or it may be caught at the outer turning point of a compact, low-angular-momentum orbit that plunges into the inner Galaxy and reaches apocentre near the Solar radius. Both share the turning-point kinematics but differ sharply in inner-Galaxy reach—the distinction the orbit modelling below sets out to quantify. We therefore compare the low-$V_{\rm grf}$ catalogue with matched Gaia DR3 control samples across higher velocity bands and treat external spectroscopic matches as context for the subset with available chemistry, not as a whole-catalogue population assignment.

Section 2 describes the sample definition, uncertainty propagation, controls, and adopted orbit model. Section 3 presents the empirical catalogue properties and chemical cross-matches. Section 4 gives the model-dependent dynamical results. Section 5 discusses the interpretation and its limits, and Section 6 summarises the conclusions.

Because this paper layers several kinds of statement, we distinguish them explicitly at the outset. *Direct Gaia observables*—sky position, parallax, proper motion, radial velocity, and the derived $V_{\rm grf}$—define the catalogue. *Threshold-probability outputs*—the per-star $P(V_{\rm grf} < 25\,{\rm km\,s^{-1}})$ and the resulting tiers—are Monte Carlo membership scores computed from those observables and their uncertainties. *Orbit-model outputs*—eccentricity, $R_{\rm peri}$, $R_{\rm apo}$, $z_{\rm max}$, energy, angular momentum, and actions—are deterministic functions of an adopted Galactic potential, and are model-dependent rather than measured. *Exploratory sensitivity products*—barred-potential central reaches, long-apocentre inner-reach and Sgr A* counts, resonance overlays, and chaos diagnostics—probe robustness and are labelled as such throughout. The conclusions rest most heavily on the first two layers. When both point-estimate and Monte Carlo orbit products are quoted, the sampling scheme is labelled explicitly; for eccentricity in particular, the deterministic point-estimate integration is systematically higher than the per-star Monte Carlo posterior medians because e is bounded above by unity.

2. SAMPLE DEFINITION AND METHODS

2.1. *Gaia DR3 Sample*

The parent source is Gaia DR3 (Gaia Collaboration et al. 2023), restricted to stars with full six-dimensional phase-space information: sky position, parallax, proper motion, and radial velocity. Radial velocities in DR3 are available for approximately 33.8 million stars observed by the Radial Velocity Spectrometer (Katz et al. 2023). This 6D subset has a non-trivial selection function in sky position, magnitude, and colour (Castro-Ginard et al. 2023; Cantat-Gaudin et al. 2023); it is not a neutral slice of the Galaxy, and distant 6D samples are commonly giant-heavy, with incompleteness expected for cool faint dwarfs.

The sample was constructed from a local mirror of the Gaia DR3 source table using the columns needed for a six-dimensional solution: `source_id`, `ra`, `dec`, `parallax`, `pmra`, `pmdec`, `radial_velocity`, the corresponding uncertainties and astrometric correlations, broad-band photometry, `ruwe`, and `duplicated_source`. The equivalent Gaia Archive selection is `radial_velocity IS NOT NULL`, `parallax > 0`, `pmra IS NOT NULL`, `pmdec IS NOT NULL`, and `parallax_over_error > 5`. No RUWE or duplicated-source cut is applied to define the point-estimate catalogue; those flags define the Gold subset below. Heliocentric observables are transformed into Galactocentric coordinates and velocities, and Galactic rest-frame speeds are computed.

To avoid imposing the final catalogue boundary during candidate construction, we first scanned the full local Gaia DR3 source mirror and retained an intentionally broad legacy buffer with preliminary $V_{\rm grf} < 200\,{\rm km\,s^{-1}}$. This buffer contains 5,867,654 unique Gaia source IDs. After the final parallax zero-point, photogeometric-distance, and Solar-frame conventions are applied, 2,755 sources have corrected point-estimate $V_{\rm grf} < 25\,{\rm km\,s^{-1}}$ within the 20,829-source near-threshold candidate pool. Monte Carlo propagation of the observables then defines the catalogue tiers used throughout the paper: Tier A ($P > 0.95$, 289 stars), Tier A+B ($P > 0.84$, 541 stars), and Tier A+B+C ($P > 0.50$, 1,952 stars). The remaining 804 corrected point-estimate sources below $25\,{\rm km\,s^{-1}}$ have $P \leq 0.50$ and are retained as Tier D rather than as part of the primary catalogue.

This is not a volume-complete sample and it is not intended as a complete inventory of all such stars in the Milky Way. It is a Gaia DR3 6D catalogue defined by a simple instantaneous kinematic state and reported with explicit threshold-membership probabilities.

To keep the role of each count unambiguous, Table 2 maps each tier and derived sample to the results it

Table 1. Sample accounting under the final distance and Solar-frame conventions. The candidate pool is the Gaia DR3 6D source list recovered from the full-parent buffered scan; the corrected point-estimate row applies the Lindegren+2021 parallax zero-point, Bailer-Jones+2021 photogeometric distances, and the default Galactocentric frame before evaluating the $V_{\rm grf} < 25\ {\rm km\,s^{-1}}$ boundary. Tiers are mutually exclusive except for the explicitly cumulative Tier A+B and Tier A+B+C rows.

Set	N
Gaia DR3 6D candidate pool	20,829
Corrected point-estimate $V_{\rm grf} < 25\ {\rm km\,s^{-1}}$	2,755
Tier A, $P(V_{\rm grf} < 25) > 0.95$	289
Tier B increment, $0.84 < P(V_{\rm grf} < 25) \leq 0.95$	252
Tier A+B, $P(V_{\rm grf} < 25) > 0.84$	541
Tier C increment, $0.50 < P(V_{\rm grf} < 25) \leq 0.84$	1,411
Tier A+B+C, $P(V_{\rm grf} < 25) > 0.50$	1,952
Tier D, point-estimate $V_{\rm grf} < 25\ {\rm km\,s^{-1}}$ and $P \leq 0.50$	804
Candidate-pool sources outside the corrected cut	18,074

anchors. The **primary catalogue** for the paper is Tier A+B ($N = 541$); orbit-derived medians and population distributions are computed on the broader Tier A+B+C sample ($N = 1{,}952$). Selection-function and chemistry diagnostics are reported only as population-scale context, never as a redefinition of the primary catalogue.

2.2. *Quality Control*

Because the slowest stars are the most vulnerable to measurement error, we report conservative high-quality subsets in addition to the tiered catalogue, following the Gaia DR3 catalogue-validation guidance (Babusiaux et al. 2023). The Gold observational-quality subset is defined inside Tier A+B by RUWE < 1.4, an acceptable DR3 RVS quality flag, parallax signal-to-noise > 10, and fractional distance uncertainty $\sigma_d/d < 0.15$. Dynamical robustness is reported separately in Section 4.5, where additional orbit-derived criteria are applied.

Operationally, "acceptable DR3 RVS quality" means the released `rv_quality` field is `ok`: the source is not flagged as `poor` by either $p(\chi^2_{\rm RV}) < 0.01$ or `rv_expected_sig_to_noise`< 2, and is not flagged as `marginal` by the large-amplitude/few-transit or faint out-of-template-temperature screens defined in the public pipeline script.

Several later diagnostics necessarily use narrower subsets. The α-classified spectroscopic subset is defined only by the availability of APOGEE or GALAH [Fe/H] and alpha-element abundances within Tier A+B+C; it is used for within-subset literature-region comparisons, not for whole-catalogue population fractions. The selection-function inverse-weighted sums are likewise reported only as observability context, because many slow stars occupy sparse colour–magnitude cells where the Gaia DR3 RVS selection-function tool returns its prior mean.

Table 4. Sensitivity of the main orbit summaries to removing stars flagged as poor by the internal Gaia DR3 RVS-quality screen. Values, including median eccentricity, are point-estimate QA diagnostics; the qualitative compact, radial-orbit result is unchanged.

Sample	N	$\widetilde{R_{\rm peri}}$ (pc)	$\tilde{e}$
Tier A+B+C, all RV qualities	1,952	116	0.964
Tier A+B+C, excluding poor RVS	1,847	115	0.964
Tier A+B, all RV qualities	541	113	0.972
Tier A+B, excluding poor RVS	508	114	0.972

2.3. *Velocity-Threshold Uncertainty*

The catalogue is defined by threshold-membership probability rather than by a hard point estimate alone. For every source in the propagated candidate pool we draw Monte Carlo realisations of $(\varpi, \mu_{\alpha*}, \mu_\delta)$ from the Gaia parallax–proper-motion covariance submatrix, holding sky position fixed because the positional uncertainties are negligible for the velocity scale considered here. We draw radial velocity independently from its quoted uncertainty, apply the Lindegren et al. (2021) zero-point correction, and draw distance from the asymmetric Bailer-Jones et al. (2021) photogeometric posterior, following the parallax-handling guidance of Luri et al. (2018). We then recompute $V_{\rm grf}$ for each realisation using the same Solar parameters and coordinate convention as the point-estimate catalogue. This explicit propagation is complementary to recent Gaia DR3 Cartesian catalogue work that compares second-order and Monte Carlo covariance propagation for the full RVS sample (Zhang et al. 2025).

Tier-fraction percentages reported below carry Beta$(k+\frac{1}{2}, n-k+\frac{1}{2})$ Jeffreys 68% intervals (Cameron 2011); integer-count diagnostics (e.g., long-apocentre inner-reach counts) carry Gehrels-style Poisson 1σ bands.

The Monte Carlo scheme uses 500 baseline realisations per source, refines stars in the transition band $0.30 < P < 0.70$ to 5,000 realisations, and assigns 10,000 realisations to stars whose point-estimate $V_{\rm grf}$ lies within $2\,{\rm km\,s^{-1}}$ of the threshold. A separate 100-star

Table 2. Which sample anchors which result. The primary catalogue for the paper is Tier A+B ($N = 541$); orbit medians and population distributions use Tier A+B+C ($N = 1{,}952$). Every numerical claim in the paper names its anchoring sample explicitly.

Sample	Used for
Tier A ($N = 289$, $P > 0.95$)	Highest-confidence empirical subset; robustness anchor.
Tier A+B ($N = 541$, $P > 0.84$)	**Primary catalogue**; Gold subsets; selection-function context.
Tier A+B+C ($N = 1{,}952$, $P > 0.50$)	Orbit-derived medians; broad catalogue characterisation.
α-classified ($N = 117$ in Tier A+B+C)	Splash/GSE/Aurora/disc fractions only.
Tier A+B SF-weighted	Population-scale context; never per-star weights.

Table 3. Sample rows from the Tier A+B primary catalogue. The full table is available in machine-readable form in the review package as `mrt/catalogue_tierAB_mrt.txt`; the FITS version is `catalogues/catalogue_expanded_tierAB.fits`.

Gaia DR3 source_id	R.A. (deg)	Decl. (deg)	$V_{\rm grf}$ ($\rm km\,s^{-1}$)	$P(V_{\rm grf} < 25)$	$N_{\rm MC}$	Tier	RVS QC
10283049355584128	50.09688	6.77727	15.64	0.920	500	B	ok
19320583963797632	39.10617	7.09534	19.11	0.968	500	A	ok
36729808698742528	57.56126	11.89217	20.89	0.908	500	B	ok
47180254402996480	61.88618	18.71729	14.48	0.994	500	A	ok
66601512401466368	58.30250	24.51245	21.86	0.884	500	B	ok
77358206452039808	32.47388	14.49229	11.33	0.972	500	A	ok
78171462803918464	29.93784	15.14448	9.80	0.960	500	A	ok
127624407740600576	37.46443	27.57781	9.24	0.892	500	B	ok

Note—Only eight rows are shown for form and content. The electronic table contains all 541 Tier A+B sources.

convergence check compares 500 and 5,000 realisations. In this release, 1,623 stars receive the 5,000-realisation refinement and 1,261 receive the 10,000-realisation near-threshold refinement. We use the resulting $P(V_{\rm grf} < 25\,{\rm km\,s^{-1}})$ values to define Tier A ($P > 0.95$), Tier B increment ($0.84 < P \leq 0.95$), Tier C increment ($0.50 < P \leq 0.84$), and Tier D (corrected point estimate below $25\,{\rm km\,s^{-1}}$ but $P \leq 0.50$).

Because every star carries a membership probability, the expected number of contaminants in any tier follows directly as $\sum(1-P)$ over its members. This gives 30.8 ± 5.3 expected false positives in the 541-star Tier A+B sample, or $94\%\pm1\%$ nominal expected purity, and 533 ± 18 in the 1,952-star Tier A+B+C sample, or $73\%\pm1\%$ nominal expected purity; the 289-star Tier A core has $99\%\pm1\%$ nominal expected purity. These uncertainties are the Bernoulli standard errors implied by the internal per-star probabilities, while the finite Monte Carlo sampling error on the summed purity is smaller than 0.05 percentage points for both the Tier A+B and Tier A+B+C samples. Tier A+B is therefore the recommended high-confidence working sample and Tier A+B+C the high-completeness extension.

The Tier A+B boundary is used for the primary catalogue because $P > 0.84$ corresponds approximately to a one-sided one-sigma probability threshold: these stars are more likely than not to remain below $25\,{\rm km\,s^{-1}}$ even after a one-sigma upward fluctuation in the propagated threshold variable. The three operating thresholds correspond to standard one-sided Gaussian-tail levels: $P > 0.95$ (Tier A) is approximately a one-sided 1.6σ threshold, $P > 0.84$ (Tier A+B) is approximately a one-sided 1σ threshold, and $P > 0.50$ (Tier A+B+C) is the more-likely-than-not boundary. Tier C is reported as the broader, lower-confidence catalogue extension rather than as part of the conservative primary sample. These Gaussian-tail correspondences are interpretive labels for the operating thresholds, not detection significances or empirical truth calibrations. We therefore describe the summed $1 - P$ values as nominal expected contamination under the internal Monte Carlo score, and test that score with both a Monte Carlo self-consistency check and a truth-labelled mock diagnostic below. Central 68% intervals on the primary numerical claims are collected in Table 22.

Because the Bailer-Jones et al. (2021) photogeometric posterior is itself conditioned on Gaia astrometry, drawing the distance independently of the astrometric covariance is an approximation. The catalogue therefore uses this Monte Carlo draw only to define threshold-membership probabilities, and treats distance-method and covariance-coupling tests as sensitivity diagnostics rather than as part of the catalogue definition.

We quantified this approximation with a stress test over the entire tier-sensitive band: all 3,342 candidates with adopted $P(V_{\rm grf} < 25\,{\rm km\,s^{-1}}) \geq 0.30$. The test keeps the BJ split-normal distance marginal but imposes a maximal Gaussian-copula anti-correlation between the parallax residual and the distance quantile, recomputing the independent- and coupled-distance probabilities at matched realisations so that ΔP isolates the coupling. The median $|\Delta P|$ is 0.009, the 84th percentile 0.021, the 95th 0.028, and the largest shift 0.055. Across the band, 114 of 3,342 sources change tier label; the full tier-label confusion matrix is given in Table 32. The primary catalogue is robust to this worst-case coupling: of the 541 Tier A+B members, 527 (97.4%) remain above $P = 0.84$, with 14 falling just below. The largest shifts occur for near-threshold sources, which the probabilistic tiering already flags (Fig. 1).

As a column-level audit of distance-prior sensitivity, we also report the Gaia parallax signal-to-noise distribution for the tiered catalogue. In the primary Tier A+B sample, 66 of 541 stars (12.2%) lie in the lower-SNR $5 < \varpi/\sigma_\varpi \leq 10$ interval and 475 (87.8%) have $\varpi/\sigma_\varpi > 10$; none have a BJ 68% fractional distance half-width above 0.15. In the broader Tier A+B+C extension, where the lower-confidence Tier C dominates, 876 of 1,952 stars (44.9%) lie in the 5–10 interval, 1,076 (55.1%) exceed 10, and 10 (0.5%) have BJ fractional distance half-width above 0.15. Thus the potentially prior-sensitive distance regime is concentrated in the completeness extension, not in the primary catalogue; the copula test above is our stress bound on how this approximation affects threshold membership.

A separate Monte Carlo self-consistency reconstruction repeats the 500-realisation baseline pass with an independent random seed, selects the 2,275

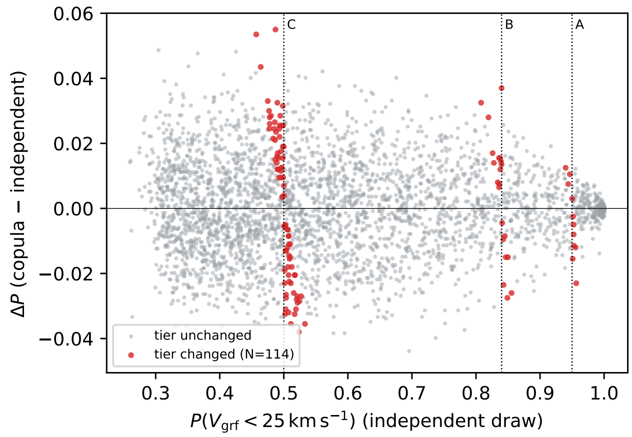

Figure 1. Change in $P(V_{\rm grf} < 25\,{\rm km\,s^{-1}})$ when the BJ-style distance draw is coupled to the Gaia parallax residual, for the full tier-sensitive band (3,342 sources with adopted $P \geq 0.30$), plotted against the independent-draw probability. The stress test imposes a maximal copula anti-correlation and probes the distance-covariance approximation in the catalogue Monte Carlo; sources that change tier label are highlighted and the tier boundaries are marked. The full tier-label confusion matrix is given in Table 32.

sources with baseline $0.30 < P < 0.70$, and compares deciles of the baseline probability against a held-out 5,000-realisation refinement. The decile medians lie close to the one-to-one relation: the median absolute decile residual is 0.0019 and the largest is 0.0040 (Fig. 2). This diagram tests numerical stability of the Monte Carlo score; it does not by itself establish empirical truth-calibration of the catalogue probabilities. It is distinct from the Gaia RVS selection-function systematic quantified separately in Section 5.

The truth-labelled GeDR3mock diagnostic provides a complementary score check. Among mock stars that enter the same RVS and parallax-SNR pipeline, binning by predicted $P(V_{\rm grf} < 25\,{\rm km\,s^{-1}})$ gives a monotonic relation between the score and the fraction of injected stars whose true speed is below $25\,{\rm km\,s^{-1}}$ (Table 5, Fig. 3). The pipeline-input Brier score is 0.031; the mock Tier A, Tier A+B, and Tier A+B+C score cuts have true sub-25 fractions of 0.997, 0.987, and 0.951, respectively. Because the mock is deliberately balanced across injected velocity bins, this is a truth-labelled score diagnostic rather than a population-prior calibration for the observed Gaia catalogue.

2.4. *Galactocentric Transformation and Supporting Orbit Model*

For the Galactocentric transformation we adopt $R_\odot = 8.178\,{\rm kpc}$ (GRAVITY Collaboration et al. 2019), a Solar

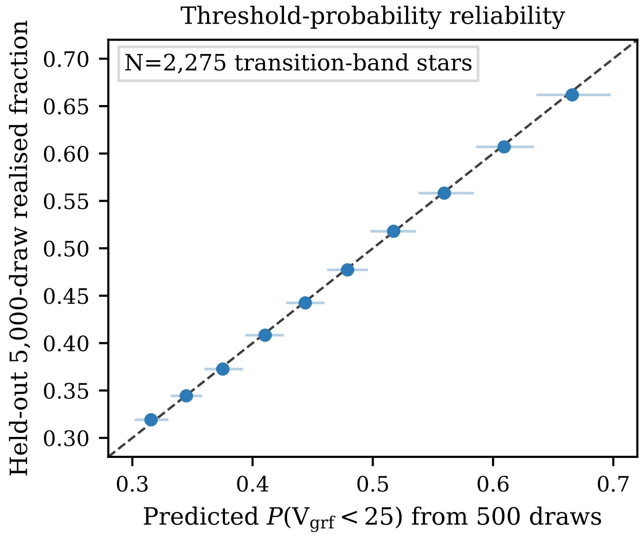

Figure 2. Monte Carlo self-consistency of the threshold-membership probabilities in the transition band. Points bin stars by the recomputed 500-realisation baseline probability; the ordinate is the realised fraction in an independent held-out 5,000-realisation refinement. Error bars are central 68% Beta$(k + \frac{1}{2}, n - k + \frac{1}{2})$ intervals (Cameron 2011).

Table 5. Truth-labelled GeDR3mock probability-score diagnostic for pipeline-input mock stars. The mock is balanced across injected true $V_{\rm grf}$ bins, so this table checks monotonic score behaviour rather than population-prior calibration. The Brier score over the pipeline-input mock is 0.031.

$P(V_{\rm grf} < 25)$ bin	N	$\langle P \rangle$	true sub-25 frac.	true$-\langle P \rangle$
0–0.10	19868	0.002	0.006	+0.004
0.10–0.30	897	0.192	0.368	+0.176
0.30–0.50	733	0.399	0.628	+0.229
0.50–0.70	843	0.602	0.805	+0.203
0.70–0.84	800	0.777	0.890	+0.113
0.84–0.95	1200	0.901	0.956	+0.055
0.95–1.00	3511	0.990	0.997	+0.007

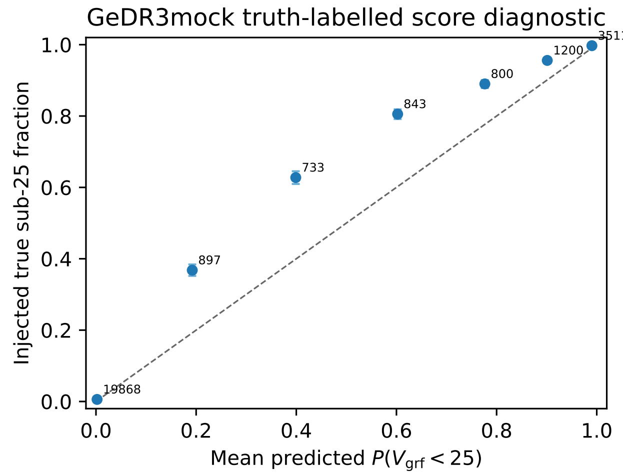

Figure 3. Truth-labelled GeDR3mock probability-score diagnostic for pipeline-input mock stars. Points bin injected stars by their predicted $P(V_{\rm grf} < 25\,{\rm km\,s^{-1}})$ and compare with the fraction whose injected true $V_{\rm grf}$ is below $25\,{\rm km\,s^{-1}}$. Numbers annotate the stars per bin. The mock prior is balanced across injected velocity bins, so the figure checks score monotonicity and high-probability behaviour rather than the population-prior calibration of the observed catalogue.

height above the plane of 25 pc, Solar peculiar motion $(U, V, W) = (11.1, 12.24, 7.25)\,{\rm km\,s^{-1}}$ from Schönrich et al. (2010), and $V_{\rm circ} = 229.0\,{\rm km\,s^{-1}}$ at the Solar radius, consistent with the Eilers et al. (2019) rotation-curve value ($v_c(R_0) = 229.0 \pm 0.2\,{\rm km\,s^{-1}}$) and with the axisymmetrised Hunter et al. (2024) potential used for the orbit integrations. The coordinate transformation is performed with `astropy.coordinates` (Astropy Collaboration et al. 2013) and the resulting Cartesian Galactocentric phase-space vectors are passed to `agama`. Three additional Solar-parameter variants from the recent literature (GRAVITY Collaboration 2022; Schönrich 2012; Reid & Brunthaler 2020) are used as a sensitivity bracket in Section 4.5.

We apply the per-star Lindegren et al. (2021) Gaia parallax zero-point correction and use the Bailer-Jones et al. (2021) photogeometric distance posterior as the default distance estimator. The inverse-parallax distance, with and without the zero-point correction, is retained only as a sensitivity diagnostic. In the propagated candidate pool, the adopted photogeometric-distance convention gives 2,755 point-estimate sources below the cut before probability-tiering.

The default supporting orbit model is the static, axisymmetrized form of the analytic Milky Way potential of Hunter et al. (2024). The underlying Hunter+2024 model is not Gaia-DR3-calibrated; it combines analytic bulge/bar, nuclear, disc, and halo components constrained by rotation-curve and gas-terminal-velocity data. We use its axisymmetrized form as the catalogue default because it keeps orbital quantities such as E, L_z, and Stäckel-fudge actions well-defined. Orbit integration and action estimates use `agama` (Vasiliev 2019); actions are computed with the Stäckel-fudge formalism (Binney 2012; Sanders & Binney 2016). Point-estimate integrations span 4 Gyr with 2,001 stored trajectory samples, while the 5,000-realisation orbit Monte Carlo uses the same 4 Gyr interval with 1,001 stored samples per realisation. This interval samples many radial periods for the compact inner-reaching orbits ($T_r \lesssim 100\,\mathrm{Myr}$ at $R_{\mathrm{peri}} \sim 100\,\mathrm{pc}$) while keeping the orbit products descriptive rather than evolutionary. A static-potential re-integration of a random 100-star Tier A subset to 8 Gyr changes the sample medians by only $+1.6\,\mathrm{pc}$ in R_{peri}, $+0.010\,\mathrm{kpc}$ in R_{apo}, and $+0.060\,\mathrm{kpc}$ in z_{max} relative to the 4 Gyr products. A rigidly rotating barred Hunter/Sormani variant (Sormani et al. 2022) is used as a sensitivity test for non-axisymmetric inner-Galaxy effects, not as the default catalogue definition. The static integrations conserve energy to a median relative drift of 6×10^{-7}; in the rotating barred frame the inertial-frame energy is not conserved (median relative drift 2%, 99th percentile 12%), but the Jacobi integral $E_J = E - \Omega_p L_z$ is conserved to a median 5×10^{-7}, confirming integration fidelity.

In this manuscript, orbit-derived quantities such as eccentricity, R_{peri}, R_{apo}, z_{max}, energy, angular momentum, and actions are presented as outputs of the adopted model. They are informative, but they remain model-dependent, especially for stars whose inferred pericentres approach the central few tens of parsecs. Pericentres, apocentres, and maximum heights are measured directly from the integrated trajectories, not inferred from Stäckel-fudge actions. The reported single-evaluation actions are accompanied by a full-sample convergent audit: for every Tier A+B+C star they are compared with orbit-time-averaged actions over the 4 Gyr integration. The azimuthal action $J_\phi \equiv L_z$ is recovered exactly and J_R to a median 8%, but Stäckel-fudge accuracy degrades for the vertical action J_z of these plunging orbits (Wright & Binney 2026): its median fractional difference is 62% over Tier A+B+C (85% in Tier A), and 1,164 of the 1,952 stars are flagged `sampled_poor`. The catalogue therefore ships orbit-time-averaged actions and a per-star action-reliability flag alongside the single-evaluation values, and we treat J_z-dependent statements as qualitative. This caveat is sharpened in Section 4.5 alongside the barred-potential sensitivity test.

2.5. *Matched-Control Sample*

To test whether orbit-derived properties vary continuously with Galactic rest-frame speed, we constructed a multi-band velocity-stratified control sample from the full Gaia DR3 6D catalogue. Four velocity bands were defined: 25–50, 50–100, 100–200, and 200–260 $\mathrm{km\,s^{-1}}$. Within each band, controls are kernel-weighted on the slow catalogue's broad observability marginals in G, $\log d$, and $\sin|b|$, with the same valid-radial-velocity requirement as the slow-star sample. The effective control samples contain 1,955, 2,586, 2,788, and 2,749 stars for the four bands respectively.

The adequacy of this weighting is audited rather than assumed. For nine observability covariates—G, $\log d$, $|\sin b|$, $\sin l$, $\cos l$, BP–RP, $\ln \mathrm{RUWE}$, $\log_{10}(\varpi/\sigma_\varpi)$, and G_{RVS}—we compute the standardized mean difference (SMD) between each control band and the slow catalogue before and after weighting (Table 6; Fig. 4). The product-of-one-dimensional observability kernel does not by itself achieve covariate balance: residual |SMD| values remain at the 0.6–0.8 level and the matched marginals are not driven toward zero, because matching each marginal independently ignores the joint covariate correlations. We therefore adopt entropy balancing (Hainmueller 2012) as the principled weighted-inference layer; it solves for the maximum-entropy control weights that match every covariate mean to $|\mathrm{SMD}| \lesssim 10^{-3}$, with band effective sample sizes of 387, 990, 1,438, and 1,474. Crucially, the monotonic increase of median pericentre with V_{grf} (§4.2) is preserved under all four weighting schemes we tested—unweighted, the observability kernel, propensity-score inverse-probability weighting, and entropy balancing—so the endpoint interpretation does not depend on the weighting choice. These reweightings use the existing local control products rather than a fresh Gaia archive scan or a replacement control-parent construction.

The 25–50 $\mathrm{km\,s^{-1}}$ band yields fewer matched controls because the candidate pool at these velocities is intrinsically small: within the Gaia DR3 6D parent sample after the basic quality requirements used for control construction (valid radial velocity and parallax signal-to-noise > 5), there are only 4,763 stars in this range, compared with 39,853 at 50–100 $\mathrm{km\,s^{-1}}$ and over 1.4 million at 100–200 $\mathrm{km\,s^{-1}}$. The extreme sparsity of the low-velocity tail—7,622 stars below 50 $\mathrm{km\,s^{-1}}$

Table 6. Covariate balance of the velocity-stratified matched controls relative to the Tier A+B+C slow catalogue, expressed as standardized mean differences (SMD). For each band the columns give the SMD before weighting (unweighted) and after entropy balancing; $|\mathrm{SMD}| < 0.1$ indicates good balance. The paper's product-of-1D observability kernel does not reduce these SMDs (Fig. 4), whereas entropy balancing matches every covariate mean to $|\mathrm{SMD}| \lesssim 10^{-3}$. The footer gives the entropy-balanced effective sample size N_{eff} and weighted median pericentre, which preserves its monotonic increase with V_{grf}.

	25–50		50–100		100–200		200–260	
covariate	before	ebal.	before	ebal.	before	ebal.	before	ebal.
G	+0.47	0.000	+0.35	0.000	+0.27	0.000	+0.26	0.000
$\log_{10}(d/\mathrm{pc})$	+0.61	0.000	+0.43	0.000	+0.34	0.000	+0.30	0.000
$\lvert\sin b\rvert$	-0.54	0.000	-0.35	0.000	-0.31	0.000	-0.35	0.000
$\sin l$	+0.03	0.000	+0.03	0.000	+0.02	0.000	+0.02	0.000
$\cos l$	+0.67	0.000	+0.44	0.000	+0.36	0.000	+0.37	0.000
$BP - RP$	+0.44	0.000	+0.33	0.000	+0.33	0.000	+0.32	0.000
ln RUWE	-0.16	0.000	-0.12	0.000	-0.06	0.000	-0.08	0.000
$\log_{10}(\varpi/\sigma_\varpi)$	-0.54	0.000	-0.33	0.000	-0.25	0.000	-0.22	0.000
G_{RVS}	+0.30	0.000	+0.23	0.000	+0.15	0.000	+0.15	0.000
N_{eff} (ebal.)	387		990		1438		1474	
med. R_{peri} (pc)	206		592		2676		3974	

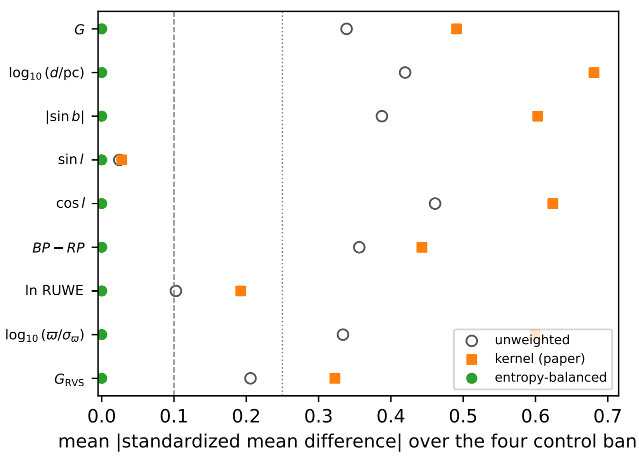

Figure 4. Covariate balance (Love plot) of the velocity-stratified matched controls relative to the Tier A+B+C slow catalogue, averaged over the four control bands. Open circles are the unweighted standardized mean differences; orange squares are the paper's product-of-one-dimensional observability kernel, which does not improve balance; green circles are entropy balancing, which matches every covariate mean to $|\mathrm{SMD}| \approx 0$. The dashed and dotted lines mark the conventional 0.1 and 0.25 balance thresholds.

out of roughly 6.8 million stars in that quality-filtered 6D parent sample, or 0.11%—shows that this low-velocity tail is sparse within the observed Gaia DR3 6D catalogue. All controls were orbit-enriched using the same static Hunter+2024 potential and integration settings as the slow-star sample.

The stratified design tests how orbital properties vary with V_{grf}, rather than comparing the slow-star sample to a single control window. The 25–50 $\mathrm{km\,s^{-1}}$ band is the adjacent kinematic comparison to the slow-star sample; the 100–200 $\mathrm{km\,s^{-1}}$ band samples a mixture of thick-disc and inner-halo kinematics; and the 200–260 $\mathrm{km\,s^{-1}}$ band brackets the circular speed at $R_\odot$ (229 $\mathrm{km\,s^{-1}}$) and therefore includes a substantial disc population.

2.6. *Catalogue Contents and Availability*

The accompanying catalogue includes Gaia `source_id`, the observables used for sample construction, adopted Galactocentric positions and velocities, V_{grf}, quality flags, and the principal orbit-derived quantities reported here (eccentricity, R_{peri}, R_{apo}, z_{max}, and related summary metrics). Orbit-derived fields are explicitly labelled as model-dependent outputs under the adopted potential rather than as direct Gaia observables. Per-star chemistry is included where available—Gaia GSP-Phot [M/H] and, for the APOGEE DR17/GALAH DR3 cross-match, spectroscopic [Fe/H], an α-abundance proxy, and a chemodynamic-population label, kept separate from the photometric metallicity—and the orbit catalogue adds both single-evaluation and orbit-time-averaged actions with a per-star action-reliability flag.

3. EMPIRICAL RESULTS

3.1. *The Sample*

The final catalogue is tiered. The Gaia DR3 6D candidate pool contains 20,829 near-threshold sources retained from a broad full-parent scan; under the final distance and Solar-frame conventions, 2,755 have corrected point-estimate $V_{\rm grf} < 25\,{\rm km\,s^{-1}}$. Point-estimate membership scales smoothly with the cut—1,420 and 4,594 sources fall below 20 and $30\,{\rm km\,s^{-1}}$ respectively—so the $25\,{\rm km\,s^{-1}}$ boundary (about 10% of the $229\,{\rm km\,s^{-1}}$ circular speed) is a convenience cut on a continuous slow tail, which is why membership is reported as probabilistic tiers rather than a hard threshold. The primary Tier A+B catalogue contains 541 stars with $P(V_{\rm grf} < 25\,{\rm km\,s^{-1}}) > 0.84$, while the broader Tier A+B+C catalogue contains 1,952 stars with $P > 0.50$. Unless otherwise stated, descriptive orbit-derived medians below are reported for Tier A+B+C and high-confidence observational-quality follow-up quantities for Tier A+B.

As a frame-dependent present-day phase-space check, we classified each star by its spherical Galactocentric radial velocity, $v_{r,{\rm GC}} = (\mathbf{r}_{\rm GC} \cdot \mathbf{v}_{\rm GC})/|\mathbf{r}_{\rm GC}|$. Negative values are inward, toward the Galactic centre, and positive values are outward. Table 7 reports this count both from the point-estimate phase-space solution and from a 5,000-realisation present-day Gaia Monte Carlo. This diagnostic uses the adopted Galactocentric frame but no orbit integration or Galactic potential. The check is included only to show that the sample does not have a large inward/outward radial-phase imbalance; the Monte Carlo probabilities are conditional on the measured catalogue and are not selection-function corrected significance tests.

Table 7. Frame-dependent present-day Galactocentric radial-phase check for the low-$V_{\rm grf}$ catalogue.

Sample	N	$N_{\rm in}$	N_0	$N_{\rm out}$	MC $f_{\rm in}$	$P_{\rm in>out}$
Tier A+B	541	242	83	216	$52.4^{+1.0}_{-1.1}$%	0.987
Gold A+B	420	196	61	163	$53.9^{+1.1}_{-1.2}$%	> 0.999
Tier A+B+C	1,952	897	259	796	$52.8^{+0.7}_{-0.7}$%	> 0.999

Note. $N_{\rm in}$, N_0, and $N_{\rm out}$ are point-estimate counts for spherical Galactocentric $v_r < -2\,{\rm km\,s^{-1}}$, $|v_r| \le 2\,{\rm km\,s^{-1}}$, and $v_r > 2\,{\rm km\,s^{-1}}$, respectively. MC $f_{\rm in}$ is the median and 16th–84th percentile interval of $N_{\rm in}/(N_{\rm in} + N_{\rm out})$ from 5,000 present-day Gaia phase-space realisations per star. This frame-dependent diagnostic uses no orbit integration or Galactic potential. $P_{\rm in>out}$ is conditional on the measured catalogue and Gaia uncertainties; it is not a selection-function-corrected significance test.

3.2. *The Primary Catalogue Survives Conservative Quality Cuts*

The Gold observational-quality subset starts from the Tier A+B primary catalogue and then applies only observational quality cuts. The catalogue Gold subset contains 420 stars after requiring RUWE < 1.4, acceptable RVS quality, parallax signal-to-noise > 10, and $\sigma_d/d < 0.15$. This gives a compact follow-up list whose definition does not depend on the orbit model.

Unresolved binaries and problematic single-epoch radial velocities can perturb Gaia RVS velocities on scales comparable to the $25\,{\rm km\,s^{-1}}$ threshold. We therefore treat RUWE and RVS-quality screens as catalogue diagnostics rather than proof of single-star centre-of-mass velocities. In the catalogue products, 28/541 Tier A+B stars ($5.2^{+1.0}_{-0.9}$%) and 60/1,952 Tier A+B+C stars (3.1 ± 0.4%) have RUWE > 1.4 (Jeffreys 68% intervals throughout); the Gold subset removes these systems, and the poor-RVS exclusion test in Section 4.5 shows that the compact radial-orbit summary is unchanged after removing flagged RVS solutions.

A direct check against the Gaia DR3 non-single-star catalogue sharpens this screen. Cross-matching by `source_id` against `gaiadr3.nss_two_body_orbit` flags 9 of the 541 Tier A+B sources (1.7%) and 23 of the 1,952 Tier A+B+C sources (1.2%) as resolved binaries—predominantly single-lined spectroscopic (SB1) solutions—which are marked in the released catalogue. Beyond these catalogued binaries, the RV-specific quality indicators bound the residual contamination (Table 8): the median DR3 RV uncertainty is $2.4\,{\rm km\,s^{-1}}$ for Tier A+B, 58% of primary-catalogue sources exceed $2\,{\rm km\,s^{-1}}$, and 6.1% have an RV-constancy χ^2 p-value below 0.01 indicative of RV variability. Because the median uncertainty is a non-negligible fraction of the $25\,{\rm km\,s^{-1}}$ threshold, the $\sigma_{\rm RV} < 2\,{\rm km\,s^{-1}}$ cut (Section 4.5) and the Gold

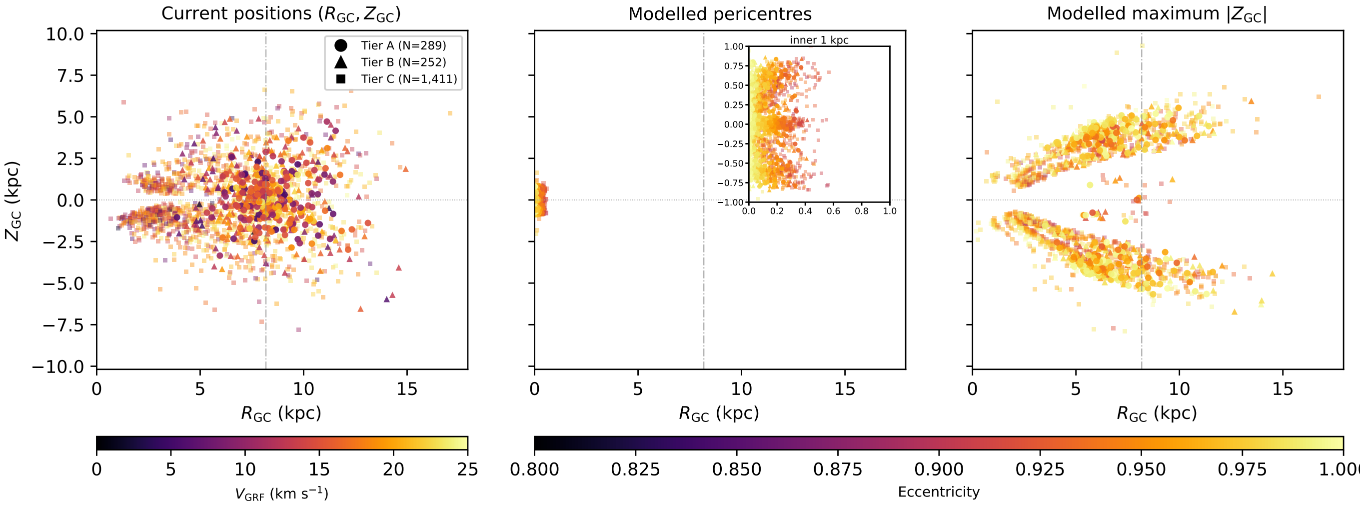

Figure 5. Edge-on orbital context for the Tier A+B+C slow catalogue (N = 1,952) in $R_{\rm GC}$ vs. $Z_{\rm GC}$. Current positions are observed coordinates, while the modelled pericentre and modelled maximum-$|Z_{\rm GC}|$ panels are model-derived locations in the adopted static Hunter+2024 potential. This broader Tier A+B+C sample is used for orbit-context visualisation; the Tier A+B primary catalogue is distinguished from the Tier C extension by marker shape. The panels show current positions, modelled pericentre positions, and modelled maximum-$|Z_{\rm GC}|$ points. Current positions are colour-coded by $V_{\rm grf}$; modelled pericentre and maximum-$|Z_{\rm GC}|$ positions are colour-coded by eccentricity, with the pericentre panel inset expanding the inner kiloparsec. The right panel shows the vertical envelope of the adopted model trajectories; apocentre radii are reported separately in Table 17 and Fig. 13. Marker shape encodes tier: circles (Tier A), triangles (Tier B), squares (Tier C). The vertical dot-dashed line marks $R_\odot = 8.178$ kpc and the Galactic centre is at $R_{\rm GC} = 0$. Because $R_{\rm GC}$ is cylindrical radius, these panels show distance from the Galactic spin axis rather than signed $X_{\rm GC}$ position. The current-position panel therefore folds signed side-of-centre information into positive radius; the pericentre panel and inset likewise show central reach, not the signed side of the Galactic centre on which each passage occurs. The signed current-position and projected-pericentre geometry is retained in the face-on view of Fig. 6.

subset are the operative defences against RV-driven false positives.

We also queried public APOGEE DR17 and GALAH DR3 radial velocities for the Tier A+B+C source IDs as a sparse external consistency audit (Table 9). The overlap is footprint-selected rather than complete: 67 Tier A+B+C stars have APOGEE RVs and 62 have GALAH RVs. The Gaia-minus-external median offsets are small ($-0.10\,{\rm km\,s^{-1}}$ for APOGEE and $+0.16\,{\rm km\,s^{-1}}$ for GALAH), with robust scatter of 1.10 and $1.72\,{\rm km\,s^{-1}}$, respectively. A minority of overlaps show $|\Delta RV| > 5\,{\rm km\,s^{-1}}$ (6.0% for APOGEE, 9.7% for GALAH), so the external check supports the bulk RV scale but does not replace follow-up confirmation for individual boundary candidates.

In this section we restrict the robustness check to empirical observables and simple descriptive properties of the sample. The corresponding effect of the Gold cut on the model-derived orbit summaries is reported separately in Section 4.5.

Table 8. Radial-velocity quality and unresolved-multiplicity audit by tier. $\sigma_{\rm RV}$ is the DR3 RV uncertainty; $f(\sigma_{\rm RV}>2)$ the fraction above $2\,{\rm km\,s^{-1}}$; $f_{\rm var}$ the fraction with $p(\chi^2_{\rm RV})<0.01$ (an RV-variability / binary indicator); $f_{\rm RUWE>1.4}$ the astrometric-binary fraction; and $N_{\rm NSS}$ the number matched in the Gaia DR3 non-single-star catalogue (`nss_two_body_orbit`).

tier	$\tilde{\sigma}_{\rm RV}$	$f(\sigma_{\rm RV}>2)$	$\tilde{N}_{\rm tr}$	$f_{\rm var}$	$f_{\rm RUWE>1.4}$	$N_{\rm NSS}$
A	2.19	54%	20	5.9%	4.2%	7
B	2.74	63%	21	6.3%	6.3%	2
A+B	2.37	58%	20	6.1%	5.2%	9
A+B+C	2.54	60%	18	5.4%	3.1%	23

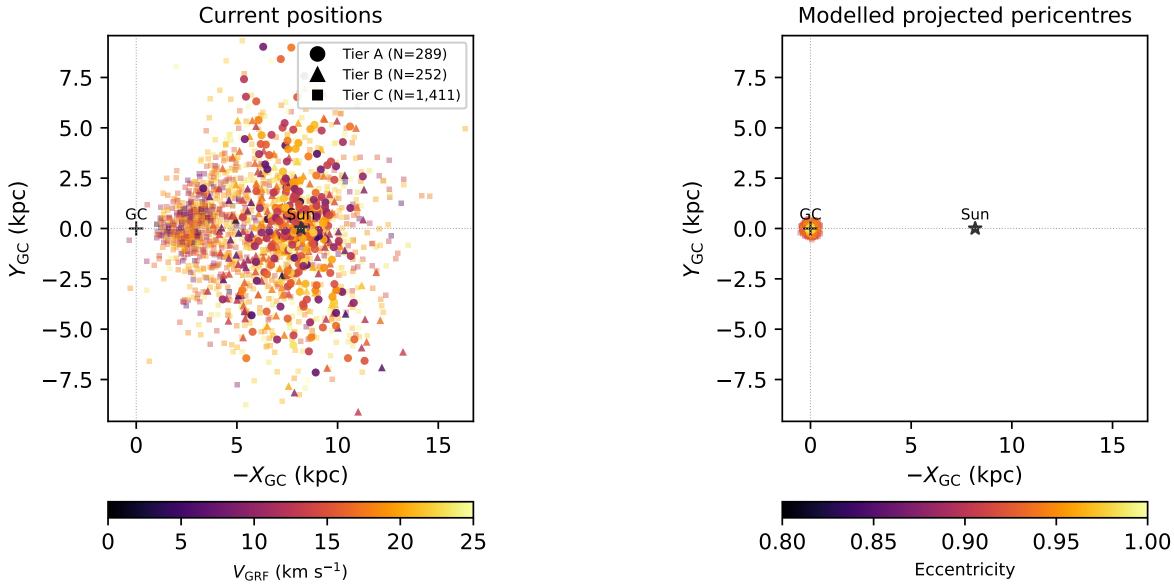

Figure 6. Face-on view of the Tier A+B+C slow catalogue (N = 1,952) in the displayed $-X_{\rm GC}$ vs. $Y_{\rm GC}$ plane, so that the Galactic centre is at left and the Sun lies to the right. As in Fig. 5, the broader Tier A+B+C orbit-summary sample is shown, with marker shape separating the Tier A+B primary catalogue from the Tier C extension. *Left:* current Galactocentric positions, colour-coded by $V_{\rm grf}$; the catalogue is observed in a Sun-centred local volume. *Right:* modelled, projected pericentre passage positions in the static Hunter+2024 potential, colour-coded by orbital eccentricity and shown on the same spatial scale as the left panel. Marker shape encodes tier (circle/triangle/square = A/B/C). The shared scale makes the inner-Galaxy reach of the catalogue directly visible: all point-estimate pericentres fall inside $R_{\rm GC} < 1$ kpc in this fine trajectory sampling, with a median Tier A+B+C pericentre of 112 pc in the 40,001-point interpolated readout.

Table 9. External radial-velocity consistency audit for the subset of Tier A+B+C stars with APOGEE DR17 or GALAH DR3 line-of-sight velocities. ΔRV is Gaia DR3 RVS minus the external survey velocity. The coverage is footprint-selected and is used only as a robustness check.

survey	$N_{\rm ABC}$	$N_{\rm AB}$	median ΔRV	robust scatter	$\lvert\Delta RV\rvert > 5$
APOGEE	67	20	-0.10	1.10	6.0%
GALAH	62	26	0.16	1.72	9.7%

Table 10. "Gold observational-quality" subset: Tier A+B stars satisfying conservative observational quality cuts only — RUWE < 1.4, RVS quality flag = "ok" (Katz et al. 2023), parallax SNR > 10, σ_d/d < 0.15, and Tier A+B membership. Dynamical assumptions do not enter this subset; it is intended for observational follow-up that does not depend on the adopted potential.

Selection criterion	Count
Tier A+B (P> 0.84) base	541
& RUWE < 1.4	513
& RVS quality = "ok"	484
& parallax SNR > 10	420
& $\sigma_d/d < 0.15$	420
Final "gold observational-quality"	**420**
Median $V_{\rm grf}$ (km s^{-1})	15.8
Median G (mag)	13.6
Median distance (pc)	2,200

3.3. *Velocity-Threshold Membership Is Probabilistic*

The corrected point-estimate catalogue contains 2,755 stars below $V_{\rm grf}$ = 25 km s^{-1}, but the threshold itself is not equally secure for every star. Monte Carlo propagation gives 289 stars in Tier A (P > 0.95), 541 stars cumulatively in Tier A+B (P > 0.84), and 1,952 stars cumulatively in Tier A+B+C (P > 0.50). The remaining 804 corrected point-estimate members form Tier D. The paper therefore uses Tier A+B as the primary catalogue and Tier A+B+C for broader population-level orbit summaries.

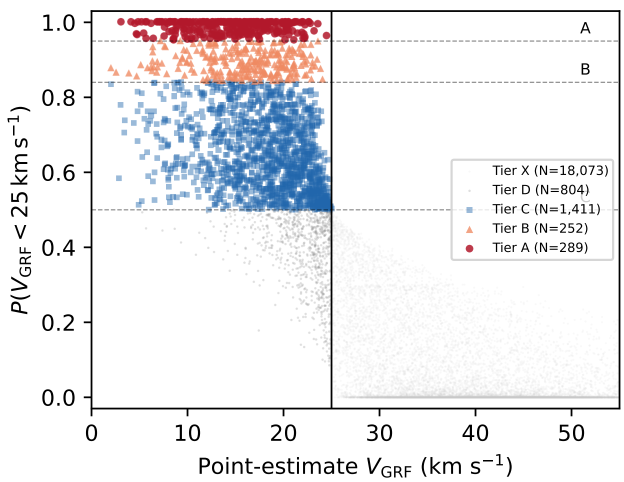

Figure 7. Velocity-threshold membership probability versus corrected point-estimate $V_{\rm grf}$ for the propagated Gaia DR3 candidate pool. The horizontal dashed lines mark the Tier A, Tier A+B, and Tier A+B+C probability boundaries; the vertical line marks the $25\,{\rm km\,s^{-1}}$ point-estimate selection boundary. The catalogue is defined by threshold probability, not by a hard point estimate alone.

As a tested observed-control smooth-tail expectation check, we compare the observed slow count against smooth velocity distributions fit to the matched-control library ($|v_{\rm GC}|$ in 25–260 $\rm km\,s^{-1}$, 10,078 stars) and extrapolated below $25\,{\rm km\,s^{-1}}$. A single trivariate-Gaussian ellipsoid, normalised to the control library, predicts $N(<\ 25\,{\rm km\,s^{-1}})\ =\ 76.4$ stars—an excess factor of 25.6 over the 1,952 observed Tier A+B+C stars (Fig. 8). A single Gaussian is, however, a weak null for a selection-truncated disc+halo velocity field, so we refit the same control velocities with a Gaussian mixture (five components by BIC) and a cross-validated kernel density (Table 12). The implied diagnostic excess is null-dependent but large for the tested nulls: 7.4× under the best-fitting mixture, 11.0× under the non-parametric KDE, and 25.6× under the single ellipsoid, each with a tight ($\lesssim$ 5%) bootstrap interval and negligible sensitivity to the $2.9\,{\rm km\,s^{-1}}$ median velocity error. We therefore treat the single-ellipsoid 25.6× as an illustrative heuristic and read the mixture-based $\sim 7\times$ as the conservative statement for the tested control-normalised smooth nulls. We do not interpret these models as a forward Gaia-selected Milky Way distribution function, nor as proof against every physically smooth but substructured velocity field; the GeDR3mock exercise below is a balanced injection-recovery test, not a population-prior forward model. This is a local comparison against the observed Gaia DR3 matched-control velocity field, not a volume-complete population estimate.

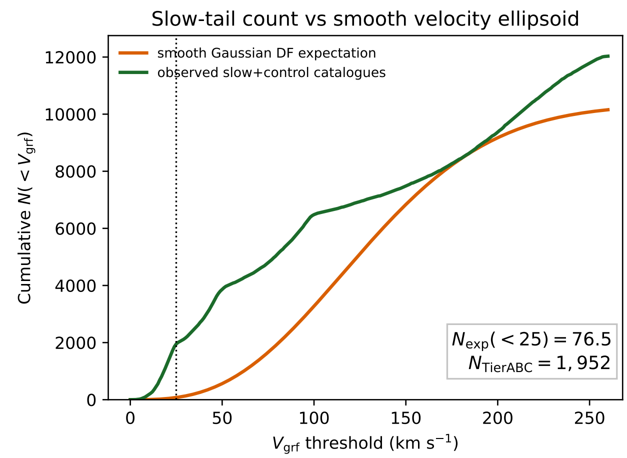

Figure 8. Observed cumulative slow-tail count compared with an illustrative smooth trivariate-Gaussian velocity ellipsoid fit to the matched-control catalogues and normalised over 25–260 $\rm km\,s^{-1}$. The excess below the adopted $25\,{\rm km\,s^{-1}}$ selection threshold motivates treating the low-$V_{\rm grf}$ catalogue as a low-velocity tail in the observed Gaia DR3 6D distribution, not as evidence for a physical discontinuity at exactly $25\,{\rm km\,s^{-1}}$ or as a complete Milky Way forward model. This single-ellipsoid baseline gives the largest of the tested smooth-null excess factors; a BIC Gaussian mixture and a cross-validated kernel density give 7.4× and 11× respectively (Table 12). Absolute pipeline normalisation is provided separately by the controlled injection-recovery test of Section 4.7.

3.4. *Chemistry Does Not Define the Sample*

Metallicity information is used here only as a diagnostic of chemical heterogeneity and estimator bias, not for population assignment. The metallicity spread in the matched subset is broad enough that the catalogue should not be treated as a single chemical population.

These official Gaia DR3 GSP-Phot metallicities are low-resolution photometric estimates with documented systematics (Andrae et al. 2023; Recio-Blanco et al. 2023): in particular, GSP-Phot tends to overestimate metallicity for cool, metal-poor giants by 0.2–0.5 dex, which biases the present sample toward solar values. In the catalogue, GSP-Phot metallicities are finite for 1,279 Tier A+B+C stars and have a median

Table 11. Monte Carlo threshold-membership probabilities for the $V_{\rm grf} < 25$ km s^{-1} cut. Astrometric realisations use the Gaia DR3 parallax–proper-motion covariance, with sky position held fixed, plus an independent radial-velocity error, with distance drawn from the asymmetric Bailer-Jones+2021 photogeometric posterior. The candidate pool contains 20,829 Gaia DR3 sources; after the final correction and distance convention, 2,755 have corrected point-estimate $V_{\rm grf} < 25$ km s^{-1}. "Sources newly admitted beyond the legacy preselection" is the count of Tier A+B+C members that do not appear in the pre-v15 slow-velocity preselection list maintained during catalogue development; they are the additions enabled by the zero-point-corrected, propagated, probability-tiered pipeline reported here.

Diagnostic	Value
Candidate pool propagated through MC	20,829
Corrected point-estimate $V_{\rm grf} < 25$ km s^{-1}	2,755
$P(V_{\rm grf} < 25$ km s$^{-1}) > 0.95$ (Tier A)	289
$0.84 < P(V_{\rm grf} < 25$ km s$^{-1}) \leq 0.95$ (Tier B increment)	252
$P(V_{\rm grf} < 25$ km s$^{-1}) > 0.84$ (Tier A+B; primary)	541
$0.50 < P(V_{\rm grf} < 25$ km s$^{-1}) \leq 0.84$ (Tier C increment)	1,411
$P(V_{\rm grf} < 25$ km s$^{-1}) > 0.50$ (Tier A+B+C)	1,952
Tier D (point-estimate < 25 km s^{-1}, $P \leq 0.50$)	804
Median $P(V_{\rm grf} < 25$ km s$^{-1})$ for Tier A+B+C	0.70
Sources newly admitted beyond the legacy preselection (Tier A+B+C)	1,230

Table 12. Implied slow-tail excess factor for the Tier A+B+C catalogue ($N = 1{,}952$ below 25 km s^{-1}) under three tested smooth velocity-distribution nulls fitted to the observed matched-control library ($|v_{\rm GC}|$ in 25–260 km s^{-1}, $N = 10{,}078$). Intervals are 16th–84th percentile bootstrap ranges over the control library. The diagnostic excess is large under every tested observed-control null, but its magnitude is null-dependent: the single ellipsoid is an illustrative heuristic and the BIC mixture gives the conservative tested value. These rows are not a forward Gaia-selected intrinsic-DF prediction.

smooth null	$N_{\rm pred}(< 25)$	excess	16–84% range
Gaussian mixture (BIC)	262.8	7.4×	7.3–7.7
Single trivariate Gaussian	76.4	25.6×	25.3–26.2
Kernel density (CV)	177.0	11.0×	10.7–11.3

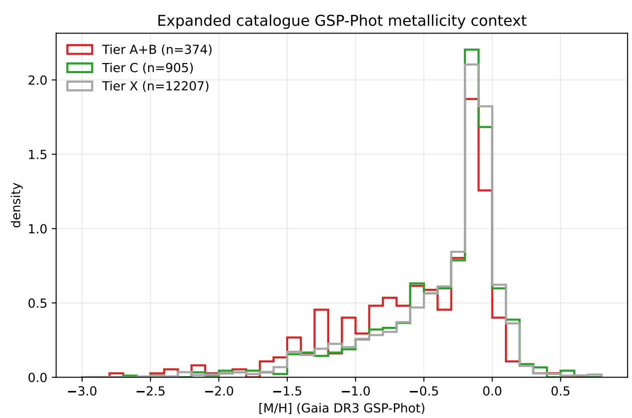

Figure 9. Gaia GSP-Phot metallicity distribution for the catalogue, shown as a low-resolution contextual diagnostic. These values are not used for population-region assignment.

[M/H] = −0.24. The spectroscopic alpha-subset discussed below is more metal-poor, with median [Fe/H] = −0.81. This offset is not interpreted as a population difference: it reflects both the known GSP-Phot metallicity systematics and the strongly non-random targeting of the APOGEE/GALAH overlap. The Zhang et al. (2023) XP catalogue remains the preferred larger photometric metallicity comparison for a future broader cross-match, but the present paper uses only the synchronized GSP-Phot and spectroscopic products as release materials.

3.5. *Chemical Properties from Spectroscopic Cross-Matching*

We cross-matched the full 20,829-source candidate pool against APOGEE DR17 (Abdurro'uf et al. 2022) and GALAH DR3 (Buder et al. 2021) using exact Gaia source identifiers rather than positional matching. The query returns 604 APOGEE matches and 489 GALAH matches across the candidate pool. Within the Tier A+B+C sample, 67 stars have usable APOGEE α-element abundances, 62 have usable GALAH abundances, and after survey-priority deduplication 117 unique Tier A+B+C stars carry the joint [Fe/H] + α-element abundance combination required for the literature-region comparisons reported in Table 13. This 117-star α-classified subset is dominated by luminous giants in the APOGEE and GALAH footprints; it is therefore a selective but useful chemical window, not a chemically complete census. The APOGEE [Mg/Fe] and GALAH [α/Fe] scales are not homogenised onto a single abundance system in this release, so the pooled classification is used only as a literature-region context flag. Survey counts are reported separately, and no whole-catalogue population fraction is inferred from the mixed-survey subset.

Within this alpha-classified subset, 51 stars fall in a Splash-like region, 23 in a Gaia-Sausage/Enceladus-like region, 22 in an Aurora-like region, 2 in a disc-like contaminant region, and 19 are not assigned to these literature regions. These fractions are reported only within the 117-star alpha-classified subset. They are not used as whole-catalogue population fractions, because the remaining Tier A+B+C stars lack the high-resolution abundance information required for that assignment in the current release products.

The broader metallicity context is still informative. Gaia GSP-Phot metallicities are available for many candidates, but the comparison in Section 3.4 shows that GSP-Phot is biased toward solar metallicity for the cool, metal-poor giants that dominate this sample. We therefore use GSP-Phot only as a low-resolution heterogeneity check and rely on spectroscopic abundances for subset-qualified population language. Photometric metallicity alone does not supply the alpha-element information needed for the Splash/GSE/Aurora comparison cuts.

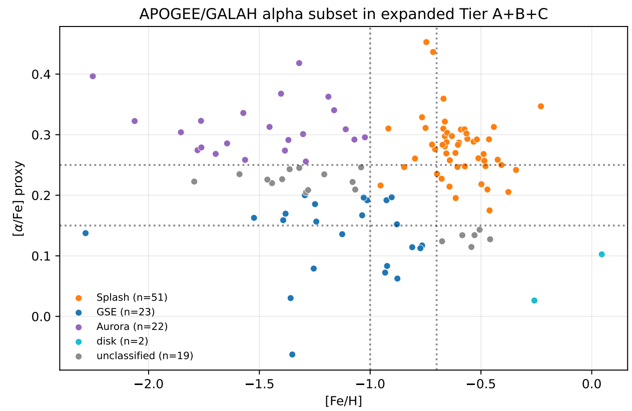

Figure 10. [Mg/Fe] (APOGEE) and [α/Fe] (GALAH) vs. [Fe/H] for the α-classified spectroscopic subset within Tier A+B+C ($N = 117$). Literature-region labels are shown only as contextual comparisons and are not treated as whole-catalogue population assignments.

Table 13. High-resolution spectroscopic chemistry available for the Tier A+B+C catalogue. APOGEE DR17 (Abdurro'uf et al. 2022) and GALAH DR3 (Buder et al. 2021) provide the subset used for α-based literature-region comparisons. Matches come from a full exact-source-id VizieR TAP cross-match against the 20,829-source candidate pool. Gaia GSP-Phot metallicities are discussed only as a biased low-resolution contextual estimator in the text, not as population assignments.

Source	N	Median [Fe/H]
APOGEE DR17, usable α	67	−0.80
GALAH DR3, usable α	62	−0.83
unique stars after deduplication	117	−0.81
Splash classified	51	−0.62
GSE classified	23	−1.04
Aurora classified	22	−1.43
disc-like contaminant	2	−0.11
unclassified in these regions	19	−1.21

Table 14. Subset-qualified chemodynamic context for the Tier A+B+C catalogue. Only stars with both [Fe/H] and [α/Fe] from APOGEE or GALAH ($n = 117$) are placed in literature comparison regions following Belokurov et al. (2020), Helmi et al. (2018), and Belokurov & Kravtsov (2022). The remaining $1{,}952 - 117 = 1{,}835$ stars are not assigned to a population in this catalogue paper. Fraction errors are Beta$(k + \frac{1}{2}, n - k + \frac{1}{2})$ Jeffreys 68% intervals on the 117-star subset (Cameron 2011); with $n = 117$ they are large, which is itself an argument against extrapolating to the full catalogue.

Population	N	Fraction	[Fe/H]
Splash	51	$43.6^{+4.6}_{-4.5}$%	−0.62
GSE	23	$19.7^{+3.9}_{-3.4}$%	−1.04
Aurora	22	$18.8^{+3.8}_{-3.3}$%	−1.43
Disc-like	2	$1.7^{+1.6}_{-0.8}$%	−0.11
Unclassified	19	$16.2^{+3.7}_{-3.1}$%	−1.21
α-classified subtotal	117	100.0%	—
not α-classified	1,835	—	—

3.6. *Sky Distribution, Observer Geometry, and Selection*

The sample is not isotropically distributed on the sky. In Galactic coordinates, 39.7 ± 1.1% of the Tier A+B+C stars lie toward the Galactic-centre direction ($l \in [330°, 30°]$), while only $6.2^{+0.6}_{-0.5}$% lie toward the anti-centre direction ($l \in [150°, 210°]$). Latitude coverage is broad, from about $-85°$ to $+87°$, with a slight southern bias in the median latitude.

This concentration is one of the strongest descriptive features of the observed catalogue. It likely reflects some combination of heliocentric observing geometry, the Gaia radial-velocity selection function, extinction and crowding effects, and the radial character of the underlying trajectories. The velocity-stratified controls are also centre-weighted, indicating that the sky distribution is partly a generic property of observed Gaia 6D stars at these distances.

The Castro-Ginard et al. (2023) Gaia DR3 RVS selection function (extending the EDR3 RVS sample-completeness framework of Everall & Boubert 2022) is therefore used as a limitation and contextual observability diagnostic for the observed catalogue rather than as a precise deprojection of the underlying Milky Way population. We evaluate the catalogue with the GaiaUnlimited DR3-RVS implementation and release the per-star selection-function table, but inverse-selection weighted counts are not used as primary population counts.

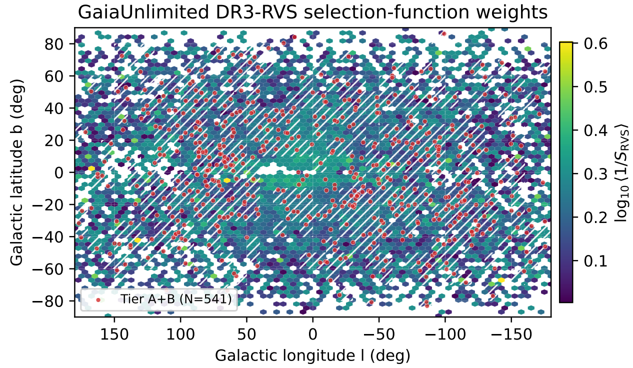

Figure 11. GaiaUnlimited DR3-RVS (Castro-Ginard et al. 2023) selection-function weight map for the expanded candidate pool, queried in the same (G_{RVS}, $BP{-}RP$, l, b) convention as the pipeline. Cell colour is the per-pixel mean $\log_{10}\langle 1/S_{\rm RVS}\rangle$; red points mark the Tier A+B primary catalogue ($N = 541$). White hatching flags pixels in which $\geq$ 50% of catalogue stars sit in GaiaUnlimited cells with parent count $n < 10$, the regime where the Beta(k+1, $n{-}k$+1) posterior is prior-dominated. The Tier A+B locus overlaps the hatched region substantially, consistent with the $f_{n<10} =$ 83.2% of stars and 88.7% of $\sum 1/S_{\rm RVS}$ reported in Table 15.

Table 15. Gaia DR3 RVS selection-function diagnostics for the catalogue. The catalogue is selected from the observed Gaia DR3 6D/RVS subset and is therefore not a volume-complete Milky Way census. $N_{\rm valid}$ is the number of sources for which the GaiaUnlimited DR3-RVS implementation (Castro-Ginard et al. 2023) returned a finite selection-function weight at the source's $(G, BP{-}RP, l, b)$ cell; the $N - N_{\rm valid}$ deficit is dominated by cells with zero parent count. $f_{n<10}$ is the fraction of $N_{\rm valid}$ sources whose underlying GaiaUnlimited cell carries a parent count below 10 (the regime in which the Beta(k+1, $n{-}k$+1) posterior remains competitive with its Beta(1, 1) prior: posterior standard error $\gtrsim 0.16$ for $n < 10$); $f_{\sum, n<10}$ is the share of $\sum 1/S_{\rm RVS}$ contributed by those prior-dominated cells. The lower block restricts each tier to the inner-Galaxy quadrant $l \in [330°, 30°]$ where the catalogue concentrates (Section 3.6). Inverse-selection weighted sums are reported only as contextual observability diagnostics and are not used as primary population counts.

Sample	N	$N_{\rm val}$	$\Sigma 1/S$	$\tilde{S}$	$f_{<10}$	$f_{\Sigma,<10}$
Full sample						
Candidate pool	20,829	19,999	37,256.3	0.50	–	–
Tier A	289	287	457.4	0.67	81.5%	87.5%
Tier B	252	249	427.6	0.50	85.1%	90.0%
Tier C	1,411	1,353	2,520.8	0.50	86.9%	90.1%
Tier A+B	541	536	885.0	0.67	83.2%	88.7%
Tier A+B+C	1,952	1,889	3,405.8	0.50	85.9%	89.7%
Inner-Galaxy quadrant, $l \in [330°, 30°]$						
Tier A (inner)	37	37	54.3	0.67	67.6%	74.6%
Tier A+B (inner)	90	89	143.1	0.67	69.7%	76.8%
Tier A+B+C (inner)	774	724	1,395.8	0.50	83.7%	86.3%

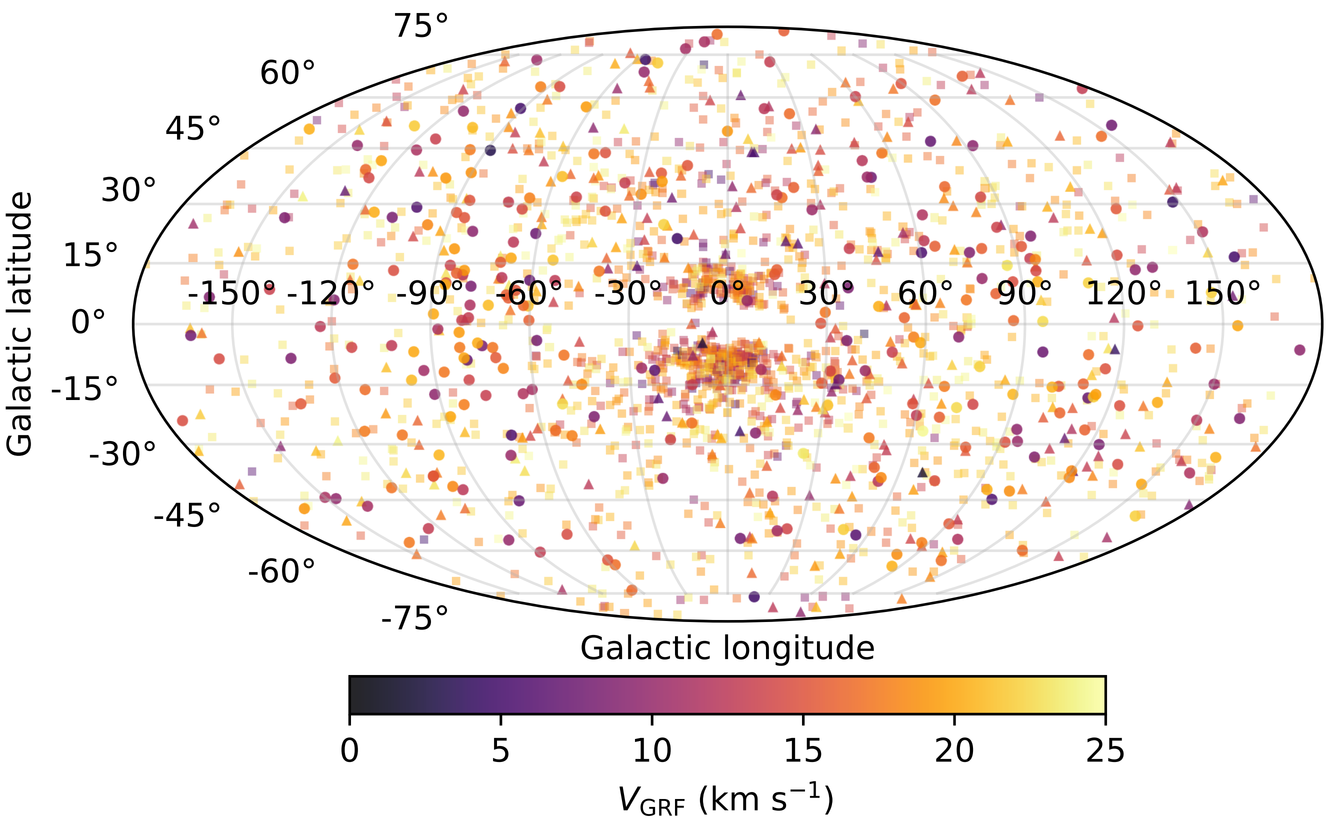

Figure 12. Aitoff projection of the Tier A+B+C slow catalogue ($N = 1{,}952$) in Galactic coordinates ($l = 0$ centred, l increasing to the left), colour-coded by V_{grf}. Marker shape encodes tier (circle/triangle/square = A/B/C). The catalogue concentrates strongly toward the inner Galaxy.

4. ORBIT-DERIVED PROPERTIES IN THE ADOPTED POTENTIAL

Within the adopted static Hunter+2024 potential, the Tier A+B+C catalogue maps onto strongly radially biased model orbits, and this result is robust across a suite of diagnostic tests. The following subsections present the primary orbit-derived properties, compare them against velocity-stratified controls, describe how the sample's properties vary with orbital scale, and assess robustness to measurement uncertainty and potential choice.

Table 16 identifies the sampling scheme behind the repeated pericentre and central-reach statistics used below. For the catalogue, the primary orbit summary uses 5,000-realisation orbit Monte Carlo propagation for every Tier A+B+C star in the adopted static Hunter+2024 potential. The velocity-stratified control comparison, halo-mass variants, barred potential variants, and central-reach counts are retained as explicitly labelled point-estimate diagnostics.

4.1. *In the Adopted Model, the Sample Is Strongly Radially Biased*

Within the adopted static Hunter+2024 potential, the orbit-derived picture is strongly radial. For the Tier A+B+C catalogue ($N = 1{,}952$), the median of the per-star posterior-median eccentricities is 0.949. All 1,952 stars have $e > 0.8$, $48.2 \pm 1.1\%$ have a posterior-median $e > 0.95$, and $0.7^{+0.2}_{-0.2}\%$ have a posterior-median $e > 0.99$ from the 5,000-realisation Monte Carlo orbit product (Table 17; Jeffreys 68% intervals).

The corresponding fractions obtained from a single deterministic point-estimate orbit integration in the same potential (Table 25) are higher: $66.7 \pm 1.1\%$ of Tier A+B+C stars have $e > 0.95$ under point-estimate integration. The difference between the Monte Carlo posterior-median fraction (48.2%) and the point-estimate fraction (66.7%) is the expected consequence of convolving the measurement-error distribution into a quantity that is bounded above by unity: the per-star posterior medians regress slightly toward lower e relative to the noise-free point estimate. Both numbers describe the same catalogue; we report the posterior-median

Table 16. Ledger for repeated pericentre and central-reach statistics. Each row refers to a different sampling scheme or sensitivity test, so the numerical values are not interchangeable. The first row is the Monte Carlo posterior-median orbit product; the control and sensitivity rows are point-estimate diagnostics unless stated otherwise.

Use in paper	Sampling scheme	Quantity reported	Interpretation
Primary orbit summary	5,000-realisation orbit Monte Carlo in the static Hunter+2024 potential	Median $R_{\rm peri} = 154\,$pc, median $R_{\rm apo} = 7.35\,$kpc, median $e = 0.949$	Uncertainty-propagated catalogue summary used for Table 17.
Control comparison	Slow-catalogue point estimate; existing controls re-weighted to catalogue observability marginals	Slow-row point-estimate medians: $R_{\rm peri} = 115.5\,$pc and $e = 0.964$; 200–260 km s^{-1} $R_{\rm peri} = 2.20\,$kpc	Supports the endpoint interpretation in Table 18.
Central reach	Point-estimate static and barred integrations	Static: 864 stars with $R_{\rm peri} < 100\,$pc; barred: 1,635	Shows that the catalogue remains strongly inner-reaching.
Halo/bar/distance sensitivity	Point-estimate halo and bar-speed reruns; BJ/parallax coupling stress test	Halo variants: median $R_{\rm peri} = 111$–$119\,$pc; bar variants: 7–31 long-apocentre inner-reach candidates	Shows the main radial-orbit result is stable while inner-Galaxy reach counts remain potential-sensitive.

fraction as the primary value because it propagates the observational uncertainties explicitly.

The same Monte Carlo orbit product gives a Tier A+B+C median $R_{\rm peri}$ of 154 pc and a median $R_{\rm apo}$ of 7.35 kpc. The maximum posterior median $R_{\rm apo}$ in the present Tier A+B+C integrations is 17.86 kpc. These values confirm that the low-speed catalogue is not dominated by ordinary near-circular disc kinematics. This compact, radial mapping is also physically expected for a catalogue selected at very low total Galactic rest-frame speed: in the adopted Galactocentric frame, such stars carry little angular momentum at the observed epoch. The robust claim is therefore the qualitative compact-pericentre, high-radiality tendency in the tested potentials, not that every quoted pericentre is an exact closest approach in the real barred inner Galaxy.

These quantities should be read as outputs of the adopted model rather than as direct Gaia observables. They are useful because they provide a coherent dynamical interpretation of the slow state. Pericentre estimates below roughly 200 pc should be treated as indicative, because the real inner Galaxy is barred and non-axisymmetric on those scales; the barred Hunter/Sormani variant in Section 4.5 brackets this sensitivity.

Crucially, the orbital *shape* is robust to the choice of axisymmetric potential. Re-integrating the catalogue in the independently calibrated, Gaia-DR3-external McMillan (2017) potential gives a near-identical point-estimate summary—median eccentricity 0.961, median $R_{\rm peri} = 123\,$pc, median $R_{\rm apo} = 7.31\,$kpc, all 1,952 stars with $e > 0.8$, and three static long-apocentre inner-reach candidates—so the radial, compact-pericentre character is not an artefact of the non-Gaia-DR3-calibrated Hunter+2024 model (Appendix A). What *is* potential-sensitive is the precise central reach and the barred long-apocentre inner-reach count, which vary by a factor of several between the static and barred families (§4.4; Appendix A). We therefore treat the high-eccentricity, compact-pericentre summary as robust and the exact inner-Galaxy reach as model-dependent.

Table 17. Summary of orbit-derived properties for the Tier A+B+C catalogue ($N = 1{,}952$) under 5,000-realisation Monte Carlo orbit propagation in the Hunter+2024 axisymmetric Galactic potential. Reported values are summaries of the per-star posterior medians; $R_{\rm peri}$, $R_{\rm apo}$, and $z_{\rm max}$ are computed in the cylindrical galactocentric frame over a 4 Gyr forward integration. Eccentricity-fraction uncertainties are Beta$(k+\frac{1}{2}, n-k+\frac{1}{2})$ Jeffreys 68% intervals (Cameron 2011); the corresponding point-estimate fractions are in Table 25.

Quantity	Value
Median posterior-median e	0.949
Posterior-median $e > 0.8$ (%)	100.0
Posterior-median $e > 0.95$ (%)	48.2 ± 1.1
Posterior-median $e > 0.99$ (%)	$0.7^{+0.2}_{-0.2}$
Median $R_{\rm peri}$ (pc)	154
$R_{\rm peri}$ p16 (pc)	89
$R_{\rm peri}$ p84 (pc)	274
Median $R_{\rm apo}$ (kpc)	7.35
Maximum $R_{\rm apo}$ (kpc)	17.86
Median $z_{\rm max}$ (kpc)	3.35

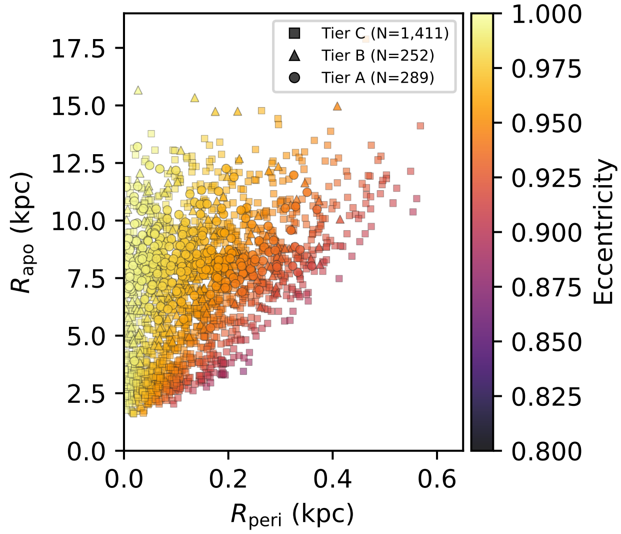

Figure 13. Pericentre vs. apocentre for the Tier A+B+C slow catalogue (N = 1,952), colour-coded by eccentricity, in the static Hunter+2024 potential. Axis ranges are set to where the adopted-model point-estimate integrations place the data: all 1,952 stars have $R_{\rm peri}$ < 1 kpc and nearly all have $R_{\rm apo}$ < 18 kpc. The long-apocentre inner-reach sensitivity test (§4.4) is reported in the text and in Table 21, not on this panel.

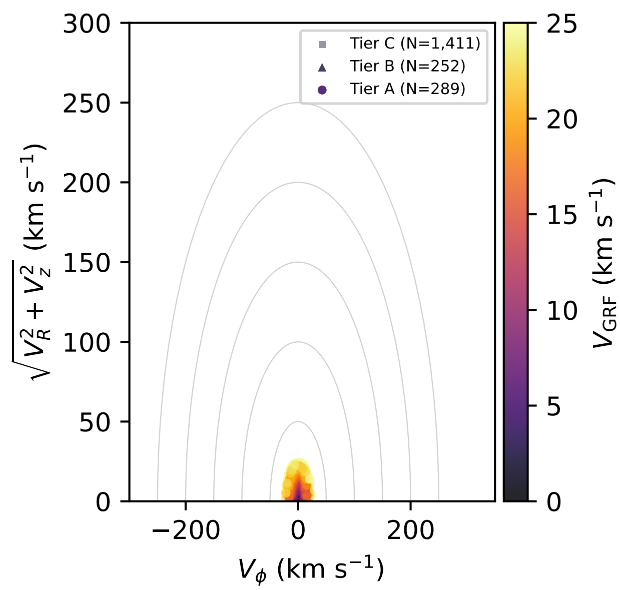

Figure 14. Toomre diagram (V_ϕ vs. $\sqrt{V_R^2 + V_z^2}$) for the Tier A+B+C slow catalogue (N = 1,952), colour-coded by $V_{\rm grf}$. The sample is concentrated at low azimuthal velocity, far from the disc locus.

4.2. *Matched-Control Comparison*

A velocity-stratified, observability-matched control comparison provides the clearest test of whether the slow-star orbit result merely reflects the typical orbital structure of Gaia stars at similar distances, or whether it varies systematically with Galactic rest-frame speed. The controls are weighted toward the slow catalogue's broad observability marginals, and the covariate balance of that weighting is audited in Section 2.5. In the catalogue, the slow-sample point-estimate median $R_{\rm peri}$ is 115.5 pc. The control-band median pericentres rise monotonically with $V_{\rm grf}$ under every weighting scheme; the exact values are method-dependent but the ordering and the slow sample's compact-pericentre endpoint are not. Under the covariate-balanced entropy weights (Table 6) the four control medians are 206 pc, 592 pc, 2.68 kpc, and 3.97 kpc; under the observability kernel they are 132.6 pc, 408 pc, 1.36 kpc, and 2.20 kpc; and unweighted they are 143 pc, 460 pc, 1.76 kpc, and 2.82 kpc. In every case the slow catalogue sits at the compact-pericentre end of a continuous velocity-stratified sequence rather than at a sharp physical boundary.

The same comparison provides a direct check on bar-resonance accumulation. The slow sample is not uniquely enhanced at the OLR: its weighted OLR fraction is 1.84%, lower than the 100–200 and 200–260 km s^{-1} controls and only modestly higher than the adjacent low-velocity controls. Barred integrations therefore remain important for individual inner-Galaxy orbits and for the long-apocentre inner-reach sensitivity test, but the matched-control comparison does not support OLR accumulation as the primary explanation for the catalogue. Figure 22 in Section 4.6 places the catalogue in L_z–J_R action space alongside the inner Lindblad, corotation, and outer Lindblad loci for the full Ω_p grid; the Tier A+B+C distribution concentrates near $L_z \approx 0$ rather than at any single pattern-speed-resonance pile-up.

The full eccentricity CDFs reinforce the same point. An unweighted two-sample Anderson–Darling comparison (Scholz & Stephens 1987) between the Tier A+B+C static-potential eccentricities and the adjacent 25–50 km s^{-1} matched-control CDF gives A^2 = 192.27 with asymptotic $p < 0.001$. The comparison to the smooth-DF mock associated with Fig. 8 gives A^2 = 36.97, again with asymptotic $p < 0.001$. Thus the near-unity eccentricity concentration is not reproduced by either the adjacent observed control band or the control-normalised smooth velocity ellipsoid. The monotonic eccentricity decline across the four control bands is likewise recovered under the covariate-balanced control weights.

Table 18. Matched-control comparison across $V_{\rm grf}$ velocity bands. Each band is integrated in the Hunter+2024 axisymmetric potential. The $R_{\rm peri}$ and e columns are deterministic point-estimate orbit summaries; the Monte Carlo posterior-median eccentricity summary is given separately in Table 17. Control rows use entropy-balanced weights that match the slow catalogue's observability covariate means (Table 6); the final column gives the older product-kernel pericentre median only as a sensitivity readout.

Sample	N	$N_{\rm eff}$	$R_{\rm peri}$	e	$R_{\rm peri,kern}$
slow A+B+C	1,952	—	116	0.964	—
25–50	1,955	387	206	0.921	133
50–100	2,586	990	592	0.796	408
100–200	2,788	1,438	2,676	0.390	1,362
200–260	2,749	1,474	3,974	0.237	2,246

Control-row labels are $V_{\rm grf}$ bands in $\rm km\,s^{-1}$; $R_{\rm peri}$ is in pc and e is dimensionless. The slow row uses the parent-complete Tier A+B+C catalogue and the point-estimate orbit product, so its $e = 0.964$ should not be interchanged with the MC posterior-median $e = 0.949$ in Table 17. Entropy-balanced effective sample sizes are rounded to the nearest star.

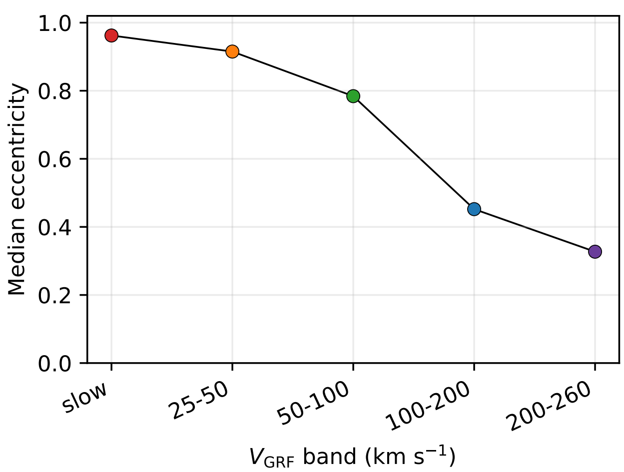

Figure 15. Median eccentricity as a function of Galactic rest-frame velocity for the Tier A+B+C slow catalogue (N = 1,952) and the four observability-matched control bands. Points show the band median eccentricities after the control bands are weighted against the slow catalogue; the full per-band eccentricity distributions are shown in Fig. 16. The monotonic decline from $e \sim 1.0$ at low $V_{\rm grf}$ to moderate eccentricity at 200–260 $\rm km\,s^{-1}$ matches the band-by-band medians in Table 18.

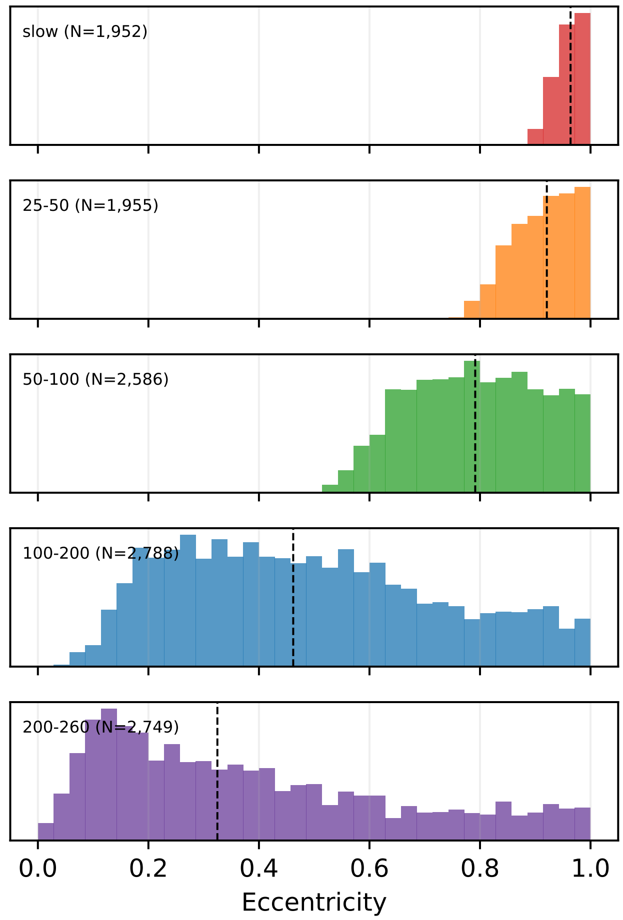

Figure 16. Eccentricity histograms for each of the five GRF velocity bands, stacked vertically. The slow Tier A+B+C band (N = 1,952) shows the extreme near-unity peak; the eccentricity distribution broadens monotonically through the four control bands toward 200–260 $\rm km\,s^{-1}$.

A dwell-time calculation in the adopted static Hunter+2024 potential helps quantify what the low-$V_{\rm grf}$ selection captures. Tier A stars spend a median 28.3% of their integrated orbital time inside $R < 2\,\rm kpc$, while Tier C stars spend 19.7% there. Conversely, the median time fraction in the solar annulus ($7 < R < 9\,\rm kpc$) rises from 11.6% in Tier A to 16.2% in Tier C. The high-confidence slow stars are therefore not merely solar-radius apocentre snapshots; they are strongly inner-reaching orbits observed during a low-speed phase.

4.3. *Dwell-Time Analysis in the Adopted Potential*

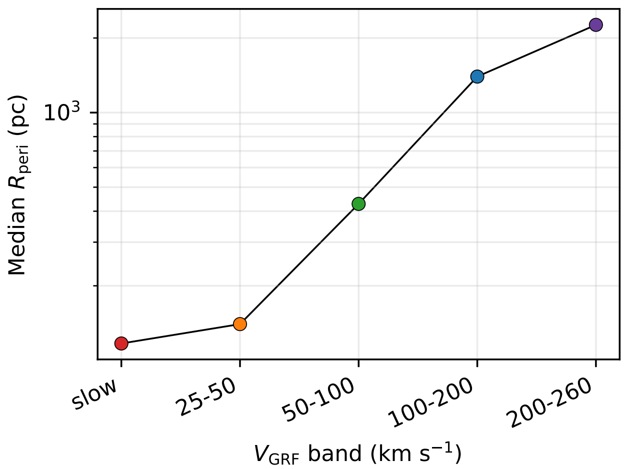

Figure 17. Median pericentre as a function of Galactic rest-frame velocity for the slow Tier A+B+C catalogue (N = 1,952) and the four observability-matched control bands. The slow band sits at the small-$R_{\rm peri}$ end of a smooth kinematic sequence; per-band pericentre medians are reported in Table 18.

Table 19. Tier-resolved orbital dwell statistics in the Hunter+2024 potential. Reported are the per-tier medians of the time fraction spent within selected cylindrical radius shells over a 4 Gyr integration, indicating that high-confidence slow-$V_{\rm grf}$ stars spend a disproportionate fraction of their trajectory inside $R < 2$ kpc despite being observed at the solar circle.

Shell	Tier A	Tier B	Tier C
Median $f(R < 1\,{\rm kpc})$ (%)	14.2	11.8	9.4
Median $f(R < 2\,{\rm kpc})$ (%)	28.3	24.1	19.7
Median $f(R < 5\,{\rm kpc})$ (%)	65.4	60.8	55.0
Median $f(7 < R < 9\,{\rm kpc})$ (%)	11.6	13.5	16.2

4.4. *Exploratory Inner-Galaxy Reach and Selection Limits*

Although the catalogue shares the defining low-$V_{\rm grf}$ state, the stars span a range of model-derived orbital scales. The Tier A+B+C catalogue has a median static-potential pericentre of 154 pc in the orbit Monte Carlo posterior medians, and all 1,952 posterior-median pericentres remain inside 1 kpc. In the dedicated point-estimate central-reach diagnostic, $R_{\rm peri}$ is the cylindrical in-plane pericentre, min $\sqrt{x^2 + y^2}$, i.e. closest approach to the Galactic spin axis rather than to Sgr A* as a point. With this definition, 864 stars have static $R_{\rm peri} < 100$ pc and all 1,952 have static $R_{\rm peri} <$ 1 kpc. This fact is visible in the right panel of Fig. 6: every star's model pericentre passage falls within the inner kpc of the Galaxy, despite the catalogue being observed in a Sun-centred local volume (left panel). In the barred sensitivity integration the corresponding cylindrical point-estimate count is 1,635 stars with $R_{\rm peri} < 100$ pc; those barred quantities are used as sensitivity diagnostics rather than as the default catalogue values.

We additionally propagated dedicated 5,000-realisation Monte Carlo closest approaches to the Galactic centre for the closest static and barred seeds: no candidate has $P(r_{\rm sph,min} < 10\,{\rm pc}) > 0.5$ and six have $P(r_{\rm sph,min} < 100\,{\rm pc}) > 0.5$, so we describe these as candidate inner-Galaxy approachers rather than a confirmed Sgr A*-passing population. The closest-approach probabilities and cumulative curves are reported in Appendix B (Table 30, Fig. 21).

The closest-approach calculation follows the same general philosophy as Gaia DR3 stellar-encounter work, where orbit integration and uncertainty propagation are used to identify robust close-passage candidates rather than to treat nominal closest approaches as definitive (Bailer-Jones 2022).

As an external source-id check, none of the 541 Tier A+B stars overlaps the 61 Gaia DR3 close-encounter entries in Bailer-Jones (2022); Table 20 reports the null cross-match explicitly.

Table 20. Overlap with the Bailer-Jones (2022) Gaia DR3 close-encounter catalogue.

Catalogue subset	BJ22 matches
Tier A+B	0

Note. BJ22 values are the median perihelion distances from VizieR catalogue J/ApJ/935/L9, Table 12. Distances in pc. The comparison uses Gaia DR3 `source_id`; this paper's values are present-day point-estimate orbit minima in the default static and barred diagnostic potentials.

We also checked the closest published low-azimuthal-velocity comparison sample, the 69 slow- and retrograde APOGEE/Gaia stars of Filion et al. (2025). All 69 resolve to unique Gaia DR3 sources and pass the present Gaia-RVS parent-source contract, but none enters Tier A+B+C under the present low-$V_{\rm grf}$ probability tiering. Three enter the 20,829-source Monte Carlo candidate pool and remain Tier X. The detailed comparison is reported in Appendix F (Table 33); it confirms that the difference is primarily one of selection definition—low V_ϕ is not the same as low total $V_{\rm grf}$—rather than an exact source-id overlap.

The radial-reach tail is also potential-sensitive. Three Tier A+B+C stars (Poisson 68% band 1.4–5.9) satisfy the long-apocentre inner-reach criterion ($R_{\rm peri} < 2$ kpc and $R_{\rm apo} > 15$ kpc) in the static Hunter+2024 integration. The barred default integration produces 21 such candidates (68% Poisson band 16.5–26.7); across the pattern-speed grid explored in Section 4.5, the count ranges from 7 to 31. Because the absolute count varies by a factor of several between the static and default-barred integrations and remains Ω_p-sensitive within the barred family, we treat these stars as bar-sensitive orbital-geometry candidates whose count is intrinsically potential-dependent, rather than as a robust population count or a merger-debris census.

Table 21. Inner-galaxy reach of the Tier A+B+C catalogue (N = 1,952). Counts are 40,001-point, parabola-interpolated point-estimate diagnostics for stars with small cylindrical in-plane pericentre $R_{\rm peri} = \min\sqrt{x^2+y^2}$, very small minimum spherical Galactocentric distance $r_{\rm sph}$, and "extreme reach" trajectories spanning the disc-to-halo radial range; the Monte Carlo orbit summary is reported separately in Table 17. The Static and Barred columns integrate under the Hunter+2024 static and barred potentials respectively; the Portail+2017 column uses the made-to-measure inner-Galaxy barred potential (Sormani-Vasiliev Agama recipe) rotated at $\Omega_p = 37.5\,\mathrm{km\ s^{-1}\ kpc^{-1}}$ to bracket the long-apocentre inner-reach count against an independent gravitational background. Candidate-row uncertainties are Gehrels (1986) Poisson 68% intervals. The barred median pericentre readout remains sampling-sensitive at tens-of-pc scales, so threshold counts are the more stable barred inner-reach diagnostic.

Diagnostic	Static	Hunter+24	Portail+17
$R_{\rm peri} < 100\,\mathrm{pc}$	864	1,635	1,635
$R_{\rm peri} < 50\,\mathrm{pc}$	467	1,504	1,494
$R_{\rm peri} < 10\,\mathrm{pc}$	99	670	637
Min $r_{\rm sph} < 100\,\mathrm{pc}$	273	784	739
Min $r_{\rm sph} < 10\,\mathrm{pc}$	10	7	5
Long-apocentre reach	$3^{+2.9}_{-1.6}$	$21^{+5.7}_{-4.5}$	$15^{+5.0}_{-3.8}$

4.5. *Robustness and Sensitivity*

The principal dynamical claims were tested against quality filtering, threshold variation, measurement uncertainty, distance estimator, Solar frame, halo mass, and barred-potential sensitivity. The full battery is collected in Appendix A: across the Gold-subset, threshold, Monte Carlo, halo-mass, barred-potential, and Portail+2017 tests, the finite-time-Lyapunov chaos diagnostic, and the Stäckel-fudge action-accuracy audit, the qualitative result is unchanged—the catalogue is compact and strongly radial, its inner-reach tail is bar-sensitive, and the high-eccentricity character survives every quality cut. Central 68% intervals on the primary numerical claims are collected in Table 22.

4.6. *Energy and Angular-Momentum Distribution*

A standard Gaia-era diagnostic is the orbital energy versus z-component of angular momentum (E–L_z); the same diagnostic underlies the substructure surveys of Naidu et al. (2020) and the proto-Galaxy mapping of Rix et al. (2022). In the static Hunter+2024 potential, the Tier A+B+C catalogue occupies low angular momentum and strongly bound inner-Galaxy phase space—the near-zero-L_z regime probed by earlier solar-neighbourhood studies (Hunt et al. 2016). This supports the same catalogue-first interpretation used throughout the paper: the sample overlaps familiar Gaia-era accretion and in-situ regions, but the present work does not assign progenitor membership from E–L_z alone. The per-star catalogue reports E, L_z, and Stäckel-fudge actions for follow-up clustering work.

The same L_z axis also locates the catalogue relative to the bar resonances. In L_z–J_R Stäckel-fudge action space the Tier A points concentrate near $L_z \approx 0$ at large J_R, away from any single corotation or outer-Lindblad pile-up, consistent with the matched-control OLR-fraction check in Section 4.2. Because the Stäckel-fudge vertical actions carry a large uncertainty for these plunging orbits (full-sample audit: median $|\Delta J_z| \approx 85\%$ in Tier A and 62% over Tier A+B+C, versus $\lesssim 10\%$ for J_R and L_z; Appendix A), we read this overlay as qualitative context rather than a quantitative resonance classification, in the spirit of recent Gaia DR3 action-space work (Palicio et al. 2023). The overlay is shown in Appendix C (Fig. 22).

4.7. *Pipeline Injection-Recovery*

To measure the direct pipeline efficiency, we injected controlled kinematic realisations into GeDR3mock sightlines, photometry, stellar parameters, and Gaia-like uncertainties (Rybizki et al. 2020). The mock contains 10,000 stars in each true-$V_{\rm grf}$ interval 0–25, 25–50, 50–100, and 100–200 $\mathrm{km\,s^{-1}}$, after the initial $G < 17$, $G_{\rm RVS} < 14$, and RVS-template-temperature cuts. Each injected star is passed through the same Lindegren zero-point, distance-posterior, 500-realisation velocity-threshold Monte Carlo, $\varpi_{\rm corr}/\sigma_\varpi > 5$, and Gaia DR3 RVS-availability stages used by the catalogue. For consistency with the v2 selection-function audit, the RVS-availability draw preserves the v1 GaiaUnlimited

Table 22. Uncertainty ledger for the primary numerical claims. Intervals are central 68% ranges: bootstrap resampling of the catalogue for the medians and excess factors, Wilson score intervals for the binomial pipeline fractions, and score-implied Bernoulli standard errors for nominal purity. Rows labelled "point" use deterministic point-estimate orbit products; the uncertainty-propagated eccentricity summary is Table 17.

quantity	value	68% interval	method
Median eccentricity (Tier A+B+C, point)	0.964	0.963–0.965	bootstrap
Median $R_{\rm peri}$ (pc, point)	115.5	112–119	bootstrap
Median $R_{\rm apo}$ (kpc, point)	7.35	7.27–7.47	bootstrap
Control 25–50 median $R_{\rm peri}$ (pc)	132.6	128–135	weighted bootstrap
Nominal purity (Tier A+B, %)	94	93–95	score Bernoulli SE
Nominal purity (Tier A+B+C, %)	73	72–74	score Bernoulli SE
Excess factor (BIC mixture)	7.4	7.3–7.7	bootstrap
Excess factor (single ellipsoid)	25.6	25.3–26.2	bootstrap
Pipeline recovery, conditional (%)	87.6	87.2–88.0	Wilson
Pipeline recovery, end-to-end (%)	60.9	60.4–61.3	Wilson
25–50 leakage fraction (%)	2.82	2.7–3.0	Wilson

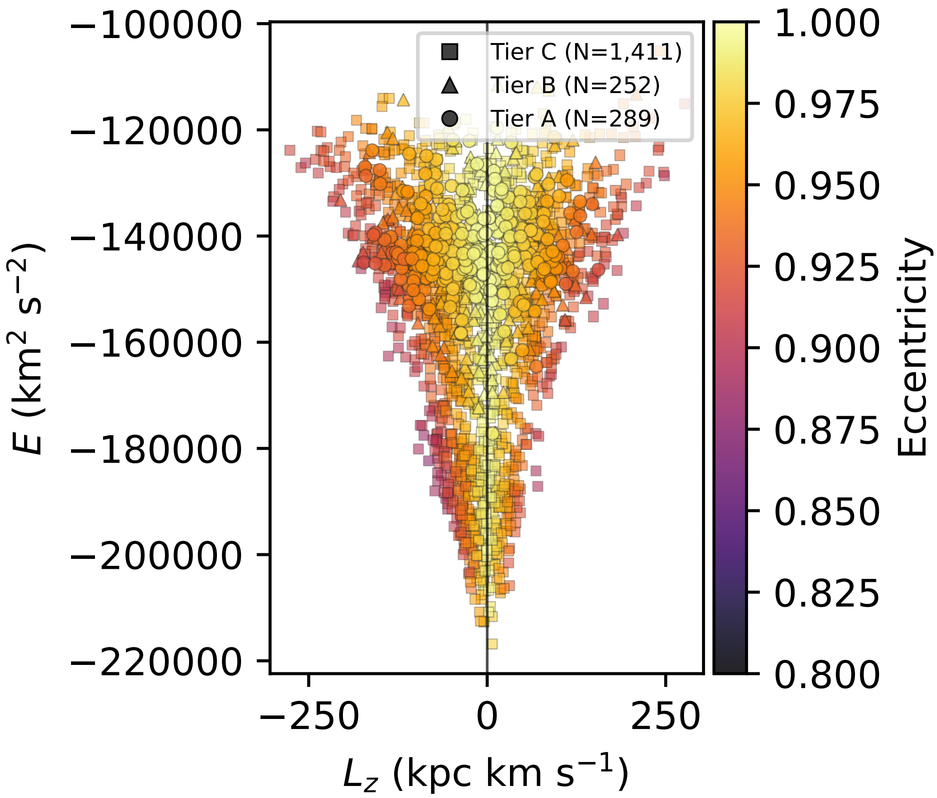

Figure 18. Tier A+B+C slow catalogue (N = 1,952) in E–L_z space, colour-coded by orbital eccentricity in the static Hunter+2024 potential. Marker shape encodes tier (circle/triangle/square = A/B/C); the vertical line marks $L_z = 0$.

convention $g = G_{\rm RVS}$ and $c = BP{-}RP$. Table 23 and Fig. 19 show that 6,086 of 10,000 true 0–25 km s^{-1} injections are recovered as Tier A+B+C, while only 282 of 10,000 stars in the adjacent 25–50 km s^{-1}

decade leak into the slow catalogue and the higher-velocity leakage is negligible. Conditional on entering the Monte Carlo after the RVS and parallax-SNR filters, the slow-bin recovery is 6,086/6,944 = 87.6%; end-to-end, including Gaia/RVS observability, it is 60.9%. Combined with the smooth-DF expectation in Section 3.3 and the selection-function prior-dominance audit in Section 5, the injection-recovery test forms the catalogue's completeness triad: the observed slow tail is large relative to the tested smooth control-normalised velocity nulls, its effective-volume scaling is limited by sparse selection-function cells, and the pipeline recovers a quantified fraction of true synthetic slow stars.

Table 23. GeDR3mock injection–recovery summary. Each true-$V_{\rm grf}$ bin contains controlled kinematic injections into GeDR3mock sightlines, photometry, stellar parameters, and Gaia-like uncertainties after the initial $G < 17$, $G_{\rm RVS} < 14$, and $3500 < T_{\rm eff} < 7000$ K observability cuts. $N_{\rm RVS}$ is the number passing a stochastic GaiaUnlimited DR3-RVS draw evaluated with the same v1 convention used for the catalogue, $g = G_{\rm RVS}$ and $c = BP{-}RP$; N_{ϖ} is the number passing $\varpi_{\rm corr}/\sigma_{\varpi} > 5$; $N_{\rm input}$ is the intersection entering the 500-realisation velocity-threshold Monte Carlo; and $N_{\rm rec}$ is the number recovered as Tier A, B, or C ($P(V_{\rm grf} < 25\,{\rm km\,s^{-1}}) > 0.5$).

True $V_{\rm grf}$ bin	$N_{\rm true}$	$N_{\rm RVS}$	N_{ϖ}	$N_{\rm input}$	$N_{\rm rec}$	$N_{\rm rec}/N_{\rm true}$
0–25	10,000	7,001	9,891	6,944	6,086	0.609
25–50	10,000	7,035	9,896	6,972	282	0.028
50–100	10,000	7,090	9,889	7,030	1	0.0001
100–200	10,000	6,966	9,879	6,906	0	0.000

5. INTERPRETATION AND LIMITS

The strongest statement in this paper is empirical. Gaia DR3 contains a tiered catalogue of stars observed at very low Galactic rest-frame speed: 541 stars at Tier A+B confidence and 1,952 stars at Tier A+B+C confidence. The exact $V_{\rm grf} = 25\,{\rm km\,s^{-1}}$ boundary is less secure for individual near-threshold sources, so the catalogue is explicitly probabilistic rather than a single hard-cut list.

A velocity-stratified control comparison across four bands shows that median pericentre grows smoothly with Galactic rest-frame speed in the adopted static potential. This continuous trend shows that the small-pericentre character of the slow-star catalogue is the low-velocity endpoint of a broader kinematic sequence in the observed Gaia DR3 6D catalogue, not a special physical boundary at exactly $25\,{\rm km\,s^{-1}}$.

The dwell-time analysis (Section 4.3) and the matched-control gradient (Section 4.2) together make a single physical statement. A kinematic cut at very low $V_{\rm grf}$ selects stars on inner-reaching, radially biased orbits during a low-speed phase. This is related to the same orbital-turning-point mechanism discussed in the accretion-debris literature, but the present catalogue does not identify the selected stars with GS/E or any other progenitor. We emphasise that this strong radial bias is in part a kinematic corollary of the selection itself: a star observed with $V_{\rm grf} \to 0$ must have near-zero angular momentum and lie close to an orbital turning point, so high eccentricity and small model pericentres are largely expected from the cut. The non-trivial findings are therefore the abundance of such stars relative to a smooth velocity field and their spatial and orbital structure, not the radial bias itself.

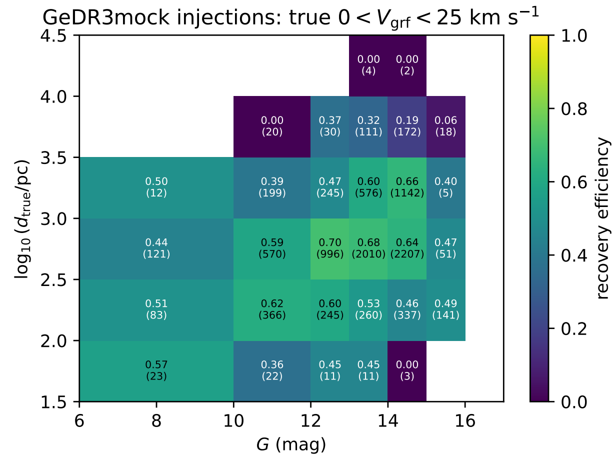

Figure 19. End-to-end recovery efficiency for the true $0 < V_{\rm grf} < 25\,{\rm km\,s^{-1}}$ GeDR3mock injections as a function of apparent G magnitude and true distance. Each cell reports the recovered fraction, with the number of injected stars in parentheses. The denominator is the post-observability injected sample; the numerator requires the RVS-availability draw, $\varpi_{\rm corr}/\sigma_{\varpi} > 5$, and Tier A+B+C recovery in the 500-realisation velocity-threshold Monte Carlo.

The point-estimate orbit integration provides the current characterisation for the main dynamical claim. All 1,952 Tier A+B+C stars have $e > 0.8$ in the default static potential, and $66.7 \pm 1.1\%$ have $e > 0.95$ under point-estimate integration (cf. the $48.2 \pm 1.1\%$ Monte Carlo posterior-median fraction in Section 4.1; the two numbers describe the same sample under different propagation choices). The strongly radial character of the sample is therefore clear in the adopted model, while posterior orbit uncertainties and potential-variant tests remain sensitivity products rather than catalogue-definition steps.

A present-day clustering null test (a DBSCAN sensitivity grid over $\varepsilon = \{5, 10, 15, 25, 50\}$ pc and `min_samples`$= \{3, 4, 5, 7, 10\}$; Ester et al. 1996; Schubert et al. 2017) returns zero compact groups for both

Tier A+B and Tier A+B+C, so the catalogue is not explained by present-day bound clusters or compact associations; this does not rule out common accretion-related origins or action-space substructure, which need not remain compact in present-day position. The grid and nearest-pair statistics are in Appendix D (Table 31).

The spectroscopic cross-match sharpens the physical picture while stopping short of an origin claim. The alpha-classified subset contains only 117 Tier A+B+C stars, so population-region fractions are quoted only inside that subset. The sample overlaps phase-space regions often discussed in the Gaia-era accretion literature (Belokurov et al. 2018; Helmi et al. 2018; Belokurov et al. 2023), and the high-eccentricity chemistry context has been studied with much larger spectroscopic samples (Lee et al. 2023). This paper is not a membership or merger-decomposition analysis. Because the sample is selected on instantaneous state, not on origin, chemistry, action space, or a fully forward-modelled selection function, we do not attempt progenitor assignment here.

As a selection-function scale diagnostic, we compute $V_{\rm eff} = \sum_i 1/p_{{\rm sel},i}$ over stars with valid GaiaUnlimited cells. For Tier A+B, excluding cells that sit exactly at the GaiaUnlimited prior mean gives $V_{\rm eff} = 423$; including those cells at the prior gives $V_{\rm eff} = 885$, a factor of 2.09 larger. For Tier A+B+C the corresponding values are $V_{\rm eff} = 1{,}512$ and 3,406, a factor of 2.25. The difference is therefore best read as a selection-function fallback systematic, not as a volume-complete population normalization.

The prior-dominated (low-parent-count, $n < 10$) fraction can be quantified directly from the underlying GaiaUnlimited (G, $BP-RP$, HEALPix) grid. For Tier A+B, 83.2% of valid stars sit in cells with parent count $n < 10$ and contribute 88.7% of $\sum 1/S_{\rm RVS}$ (Table 15; Fig. 11); restricting to the inner-Galaxy quadrant $l \in [330°, 30°]$ where the catalogue concentrates lowers this to 69.7% of stars and 76.8% of $\sum 1/S_{\rm RVS}$ – still majority prior-dominated. The choice of selection-function estimator is a smaller effect. Combining the Cantat-Gaudin et al. (2023) DR3 parent-completeness map with the Castro-Ginard et al. (2023) RVS subsample SF leaves $\sum 1/S_{\rm RVS}$ for Tier A+B unchanged at 885 to within 0.1% because the DR3 parent completeness saturates at the catalogue's $G_{RVS} < 14$ magnitude range. Substituting the older EDR3 RVS SF of Everall & Boubert (2022) yields a nominally $\sim 18\times$ larger $\sum 1/S$, but that estimator targets a smaller, brighter RVS subsample than DR3 RVS and is not a like-for-like replacement. We therefore retain the prior-dominance audit as the operative SF systematic for the catalogue.

The principal open questions are interpretive rather than catalogue-level. Spectroscopic coverage is incomplete. Very small inferred pericentres and long-apocentre inner-reach counts are model-sensitive. The sample is not volume-complete, and local halo samples cannot in general be extrapolated to the global halo without strong modelling assumptions (Sharpe et al. 2024). These limitations set the boundary of the present paper's claims.

6. CONCLUSIONS

We present a tiered Gaia DR3 catalogue of stars with very low Galactic rest-frame speeds. The primary Tier A+B catalogue contains 541 stars with $P(V_{\rm grf} < 25\,{\rm km\,s^{-1}}) > 0.84$; the Tier A high-confidence core contains 289 stars with $P > 0.95$, and the broader Tier A+B+C catalogue contains 1,952 stars with $P > 0.50$.

In the default static Hunter+2024 potential, the Tier A+B+C catalogue maps onto strongly radially biased model orbits with median eccentricity 0.949, median $R_{\rm peri} = 154\,{\rm pc}$, and median $R_{\rm apo} = 7.35\,{\rm kpc}$ from the 5,000-realisation orbit Monte Carlo.

The catalogue is not a re-statement of the local velocity ellipsoid. As an observed-control diagnostic, tested smooth velocity-distribution nulls fit to the 25–260 $\rm km\,s^{-1}$ control library and extrapolated below 25 $\rm km\,s^{-1}$ under-predict the observed slow count by a factor of 7–26 depending on the null (76.4 stars and an illustrative $25.6\times$ for a single Gaussian ellipsoid; $7.4\times$ for a BIC Gaussian mixture; Section 3.3, Table 12), and unweighted Anderson–Darling tests reject the catalogue's eccentricity distribution against both the adjacent 25–50 $\rm km\,s^{-1}$ matched control and the smooth-DF mock at $p < 0.001$.

The barred-potential sensitivity sweep produces 7–31 long-apocentre inner-reach candidates across pattern speeds Ω_p = 24, 28, 33, 37.5, and 41 $\rm km\,s^{-1}\,kpc^{-1}$, while the static default produces three. An independent M2M bar (Portail et al. 2017) integrated at $\Omega_p = 37.5\,{\rm km\,s^{-1}\,kpc^{-1}}$ yields 15 such candidates and the same compact, high-eccentricity barred-orbit character as the Hunter+24 barred reference; the Gehrels Poisson 1σ intervals for the two counts overlap (Section 4.5, Table 21). Because the barred median pericentre readout is still sampling-sensitive at 40,001 points, the robust comparison is the overlapping long-apocentre inner-reach count and stable point-estimate cylindrical inner-reach population rather than a precise barred median. This cylindrical $R_{\rm peri}$ count is distinct from the Sgr A* analysis, where the six soft approachers are defined by Monte Carlo posterior probability in spherical Galactocentric distance. The long-apocentre inner-reach count is therefore robust to swapping the disc and halo background at fixed bar parameterisation.

A finite-time Lyapunov diagnostic in the barred potential shows 1/3 of the static long-apocentre inner-reach candidates, 0/10 Sgr A*-approacher seeds, and 4/100 Tier A control stars in the order-unity strongly chaotic regime (Section 4.5, Fig. 20). Bar-induced chaos is present in this sample but bounded; the orbital shape statistics it underwrites are descriptors rather than selection-defining. A direct check confirms this. Extending the convergent comparison to all 1,952 Tier A+B+C stars—catalogue Stäckel-fudge actions versus orbit-time-averaged actions over the 4 Gyr integration—gives median absolute fractional differences of 8.3% in J_R, $\sim 3 \times 10^{-8}$ in J_ϕ, and 61.8% in J_z (9.3% and 85.1% for J_R and J_z in Tier A). By the per-star reliability flag, 219 stars are `sampled_ok` ($\leq 15\%$), 569 `sampled_caution` ($\leq 50\%$), and 1,164 `sampled_poor` ($> 50\%$); the catalogue ships both the single-evaluation and orbit-time-averaged actions.

Selection-function weighting is reported as context, not as a precise deprojection. Quantitatively, 83.2% of the primary catalogue's valid sources sit in GaiaUnlimited cells at parent count $n < 10$, contributing 88.7% of $\sum 1/S_{\rm RVS}$ (Section 3.6, Table 15), and the corresponding effective-volume inflation between the prior-excluded and prior-included sums is 2.09× for Tier A+B and 2.25× for Tier A+B+C. A controlled GeDR3mock injection-recovery exercise (Rybizki et al. 2020) recovers 60.9% of true 0–25 $\mathrm{km\,s^{-1}}$ injections end-to-end and 87.6% conditional on entering the pipeline, with leakage from 25–50 $\mathrm{km\,s^{-1}}$, 50–100 $\mathrm{km\,s^{-1}}$, and 100–200 $\mathrm{km\,s^{-1}}$ of 2.82%, 0.01%, and 0% respectively (Section 4.7, Table 23). Smooth-DF excess, prior-dominance fraction, and pipeline recovery together form a three-leg quantitative defence of the catalogue's reality as more than a thresholding artefact.

A DBSCAN sensitivity grid over $\varepsilon \in \{5, 10, 15, 25, 50\}$ pc and `min_samples`$\in \{3, 4, 5, 7, 10\}$ returns zero compact groups for both the Tier A+B and Tier A+B+C samples (Table 31). The catalogue is not explained by one or more present-day bound clusters or compact associations.

Chemistry is subset-qualified. Only 117 Tier A+B+C stars carry the joint [Fe/H] and alpha-element abundance combination required for literature-region comparisons, so Splash, Gaia-Sausage/Enceladus, Aurora, and disc-like fractions are quoted only within that alpha-classified subset.

The natural next catalogue step is Gaia DR4: applying the same probabilistic low-$V_{\rm grf}$ selection to the longer-baseline release will test the stability of the present DR3 sample, retire the prior-dominated $(G, BP{-}RP)$ cells that currently set the dominant selection-function systematic, and enlarge the accessible low-speed catalogue.

The author acknowledges the use of public data from Gaia DR3 and from the spectroscopic resources cited in this paper.

AI Assistance Disclosure.—AI language-model tools were used during preparation of this manuscript for coding assistance, editorial revision, literature-navigation support, and critical feedback on draft text. These tools were not used as authors and were not treated as scientific authorities. In particular, no AI tool was used to generate, run, or independently validate the numerical analyses on which the catalogue rests: the Gaia DR3 sample construction, the Monte Carlo uncertainty propagation, the `agama` orbit integrations, the Galactocentric and threshold-membership calculations, the matched-control reweighting, the selection-function diagnostics, the statistical tests, and all figures and tables were produced and verified directly from the documented release-stage pipeline and re-executed by the author. All numerical results, validation steps, methodological decisions, interpretations, references, and final prose were reviewed and verified by the author, who takes sole responsibility for the scientific content.

Data Availability.—The observational source catalogues used in this work are publicly available from the respective survey archives (Gaia DR3 via the ESA Gaia Archive; APOGEE DR17 via the SDSS DR17 archive; GALAH DR3 via the GALAH Survey website; the Queiroz et al. 2023 StarHorse value-added catalogue via VizieR; and the Zhang et al. 2023 Gaia XP parameter catalogue via the GAVO Heidelberg TAP service as `xpparams.main`). The propagated candidate pool, the Tier A, Tier A+B, Tier A+B+C, and Tier D catalogues, the four matched-control band catalogues, the per-star Monte Carlo results, and the orbit-derived columns are archived in Zenodo at https://doi.org/10.5281/zenodo.20116134. The release includes the FITS catalogues, control catalogues, validation products, Gaia Archive selection query, pipeline scripts, potential configuration files, and environment metadata needed to reproduce the numerical products from the released intermediate tables.

Software Availability.—The review-stage analysis pipeline, including sample construction, control-band matching, orbit integration, Monte Carlo propagation, and the figure- and table-generation scripts, is available in a version-controlled GitHub repository at github.com/wodvik/gaia-slow-vgrf-catalogue. The repository also contains the review-stage catalogue products, column documentation, environment file, configuration file, Gaia Archive selection query, and the current manuscript draft.

APPENDIX

A. ROBUSTNESS AND SENSITIVITY BATTERY

The principal dynamical claims are tested here against quality filtering, threshold variation, measurement uncertainty, distance estimator, Solar frame, halo mass, and barred-potential sensitivity.

Gold-subset dynamical robustness in the adopted model.— The Gold dynamical subset starts from the Tier A+B primary catalogue and applies the observational Gold cut before comparing point-estimate orbit summaries. The Gold cut barely changes the orbit distribution: median $R_{\rm peri}$ is 121 pc in the Gold subset versus 120 pc in the full Tier A+B catalogue, and all Gold stars have finite action solutions in the adopted static potential. The subset is intended for individual-star follow-up rather than for redefining the full catalogue.

Table 24. Orbit-derived properties before and after the observational Gold cut. The Gold subset is defined in Table 10; quantities here are point-estimate outputs of the adopted static potential, including eccentricity, and are not used to define the subset. Eccentricity-fraction errors are Beta$(k+\frac{1}{2}, n-k+\frac{1}{2})$ Jeffreys 68% intervals (Cameron 2011).

Quantity	Tier A+B	Gold obs.
N stars	541	420
Median eccentricity	0.972	0.971
Fraction with $e > 0.8$ (%)	100.0	100.0
Fraction with $e > 0.95$ (%)	$81.0^{+1.6}_{-1.7}$	$79.8^{+1.9}_{-2.0}$
Fraction with $e > 0.99$ (%)	$19.0^{+1.7}_{-1.6}$	$17.9^{+1.9}_{-1.8}$
Median $R_{\rm peri}$ (kpc)	0.1125	0.1148
Median $R_{\rm apo}$ (kpc)	8.17	8.09
Median $z_{\rm max}$ (kpc)	3.92	3.85
Finite action solutions	541	420

Threshold sensitivity within the current local sample.— The $25\,{\rm km\,s^{-1}}$ threshold is a practical operating cut, not a special physical boundary. The candidate pool is therefore released with per-star $P(V_{\rm grf} < 25\,{\rm km\,s^{-1}})$ values and tier labels rather than only a hard threshold flag; alternative threshold cuts can be reconstructed directly from the machine-readable catalogue.

Table 25. Threshold sensitivity. Counts at varying $V_{\rm grf}$ cuts and the corresponding orbital eccentricity distribution under the point-estimate Hunter+2024 orbit integration. The $V_{\rm grf} < 25\,{\rm km\,s^{-1}}$ threshold used for the primary catalogue is a practical operating cut, not a sharp physical boundary. Fraction errors are Beta$(k+\frac{1}{2}, n-k+\frac{1}{2})$ Jeffreys 68% intervals (Cameron 2011). Point-estimate fractions are systematically higher than the Monte Carlo posterior-median counterparts in Table 17 because of measurement-error convolution.

Cut (${\rm km\,s^{-1}}$)	$N_{\rm point}$	$N_{\rm ABC}$	$\tilde{e}$	$e > 0.95$ (%)
< 10	188	182	0.986	$100.0_{-0.5}$
< 15	588	553	0.978	$90.4^{+1.2}_{-1.3}$
< 20	1,420	1,255	0.970	76.7 ± 1.2
< 25	2,755	1,952	0.964	66.7 ± 1.1

Velocity-threshold Monte Carlo propagation.—Adaptive Monte Carlo propagation was run for all 20,829 near-threshold candidates. The baseline pass used 500 realisations per source, with 5,000-realisation refinement in the transition region and 10,000-realisation refinement near the hard $25\,{\rm km\,s^{-1}}$ point-estimate boundary. The resulting tier labels, rather than a single point estimate, define the catalogue.

Table 26. Adaptive Monte Carlo velocity-threshold propagation for the candidate pool. Each star is sampled from the Gaia DR3 parallax–proper-motion covariance, with sky position held fixed, an independent radial-velocity error, and the Bailer-Jones+2021 photogeometric distance posterior. Transition-band stars are refined to 5,000 realisations; near-threshold point estimates are refined to 10,000. The table reports the actual number of stars in each adaptive branch.

Quantity	Value
Candidate pool	20,829
Tier A+B+C after MC tiering	1,952
Pass-1 realisations / star	500
Transition-band refinements	1,623
Near-threshold ultra refinements	1,261
Refinement realisations / star	5,000
Ultra realisations / star	10,000
Minimum / maximum recovered $P(V_{\rm grf} < 25\ {\rm km\,s^{-1}})$	0.0 / 1.0

Halo-mass sensitivity on the full Tier A+B+C catalogue.— A full-sample halo-mass sensitivity study was repeated for the Tier A+B+C catalogue, bracketing recent Milky Way mass estimates (e.g. Cautun et al. 2020; Wang et al. 2020). Varying the halo mass by $\pm 15\%$ changes the median point-estimate pericentre from 116 pc in the matched 40,001-point default integration to 119 pc and 111 pc, respectively, and leaves the long-apocentre inner-reach count unchanged at three. The high-eccentricity result is therefore driven primarily by the observed phase-space configuration: all 1,952 Tier A+B+C stars have $e > 0.8$ in the default static potential.

Bar-perturbation sensitivity for a stratified compact-orbit subset.—Because many catalogue stars reach the inner Galaxy, we also integrate a barred Hunter/Sormani variant (Sormani et al. 2022) with pattern-speed sensitivities at $\Omega_p = 24$, 28, 33, 37.5, and 41 $\mathrm{km\,s^{-1}}$ kpc $^{-1}$. Barred integrations leave the qualitative low-$V_{\rm grf}$ radiality intact but change the central reach and long-apocentre inner-reach counts. The static potential yields three such candidates, while the barred default yields 21 in the point-estimate integration. The full-sample pattern-speed sensitivity gives 8 such candidates at $\Omega_p = 24\,\mathrm{km\,s^{-1}\,kpc^{-1}}$, 7 at $28\,\mathrm{km\,s^{-1}\,kpc^{-1}}$, 8 at $33\,\mathrm{km\,s^{-1}\,kpc^{-1}}$, and 31 at $41\,\mathrm{km\,s^{-1}\,kpc^{-1}}$. The lower end directly tests the slower $\Omega_p \approx 24 \pm 3\,\mathrm{km\,s^{-1}\,kpc^{-1}}$ bar reported by Horta et al. (2024) from joint APOGEE–Gaia constraints, while the higher end brackets the commonly adopted Milky Way bar pattern-speed range around 35–45 $\mathrm{km\,s^{-1}\,kpc^{-1}}$ (Clarke et al. 2019; Zhang et al. 2024). At fixed $\Omega_p = 37.5\,\mathrm{km\,s^{-1}\,kpc^{-1}}$, varying the bar angle from 20° to 35° gives 40,001-point median $R_{\rm peri}$ readouts of $\lesssim 13$–$\lesssim 15$ pc and $N_{\rm bridge} = 17$–22; flattening the halo over $q_z = 0.80$–1.05 gives median $R_{\rm peri} = 112$–115 pc and leaves $N_{\rm bridge} = 3$. We therefore retain the barred integrations as systematic tests of inner-Galaxy geometry, not as replacements for the static-potential catalogue values. The barred median pericentre readout itself is still sampling-sensitive over 20,001–40,001 points (20 pc to 14 pc in the default barred case), whereas $N(R_{\rm peri} < 100\,\mathrm{pc})$ and this radial-reach count are stable; we therefore use the barred medians only as resolution-limited diagnostics and rely on threshold counts for robust inner-reach comparisons. This caution is motivated by recent work showing that the Galactic bar can generate or reshape halo phase-space structure, including small-pericentre and chevron-like features in Gaia DR3 (Dillamore et al. 2023, 2024), and that the bar induces measurable chaos in the orbits of nearby halo stars on the very $R_{\rm peri}$ scales sampled by this catalogue (Woudenberg & Helmi 2025). We therefore computed Agama's normalized finite-time Lyapunov indicator, $\lambda T_{\rm orb}$, in the default barred potential at $\Omega_p = 37.5\,\mathrm{km\,s^{-1}\,kpc^{-1}}$ for the three static-potential long-apocentre inner-reach candidates, the ten Table 30 Sgr A* approacher seeds, and a random 100-star Tier A control sample. The medians are 0.49, 0.47, and 0.44, respectively; none of the three long-apocentre inner-reach candidates, one of the ten Sgr A* seeds, and four of the 100 controls reach the order-unity regime that Agama flags as strongly chaotic. Figure 20 shows that the closest-approach seeds are not separated from the control sample by the chaos indicator, so the barred central-reach tail is potential-sensitive but not dominated by a uniformly high-chaos subset.

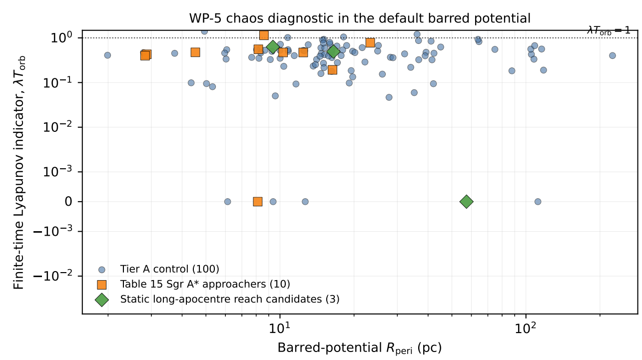

Figure 20. Finite-time Lyapunov indicator in the default barred Hunter/Sormani potential at $\Omega_p = 37.5\,\mathrm{km\,s^{-1}\,kpc^{-1}}$, plotted against the barred-potential cylindrical pericentre. The horizontal dotted line marks $\lambda T_{\rm orb} = 1$, the order-unity level used by Agama as a practical strongly chaotic threshold.

As a second-potential confirmation that the long-apocentre inner-reach result is not an artefact of the Hunter+2024 disk/halo background, we repeated the Tier A+B+C orbit integration in the made-to-measure inner-Galaxy bar of Portail et al. (2017), constructed via the analytic multi-component Agama recipe of Sormani et al. (2022) and rotated at the matching $\Omega_p = 37.5\,\mathrm{km\,s^{-1}\,kpc^{-1}}$ and 25° bar angle. The Portail+2017 potential uses an independent disk and a flattened Einasto dark halo and therefore samples a different overall gravitational landscape from the Hunter+2024 background. The same 1,952-star catalogue yields a Portail+2017 median $R_{\rm peri} \lesssim 14$ pc in the 40,001-point readout, median $R_{\rm apo} = 8.38$ kpc, median eccentricity 0.996, and $N_{\rm reach} = 15^{+5.0}_{-3.8}$ (Gehrels 68%); the central-reach counts are also reported in Table 21. This radial-reach count overlaps the default barred Hunter+2024 value of $21^{+5.7}_{-4.5}$ within Gehrels 1σ, and the median orbital shape statistics have the same high-eccentricity character, demonstrating that the ∼tens-of-candidates result and the high inner-reach concentration are robust to swapping out the surrounding disc/halo at fixed bar parameterisation.

For the same 100 Tier A control stars, we compared the catalogue Stäckel-fudge actions with actions re-evaluated along the 4 Gyr static-potential trajectories and averaged over time. The median absolute fractional differences are 8.9% in J_r, negligible in J_ϕ ($< 10^{-4}$%), and 85.1% in J_z. The near-exact recovery of J_ϕ is the expected angular-momentum check in the axisymmetric static potential, while the large vertical-action scatter quantifies the Stäckel-fudge caveat for compact plunging trajectories. We therefore keep catalogue actions as follow-up descriptors rather than as selection-defining quantities.

Radial-velocity uncertainty robustness.—As an additional defensive cut, we required Gaia DR3 radial-velocity uncertainty $\sigma_{RV} < 2\,\mathrm{km\,s^{-1}}$ on top of the Gold filter. This removes the noisier RVS solutions and retains 171 of the 420 Gold stars ($40.7^{+2.4}_{-2.4}$%). The comparatively aggressive reduction reflects that DR3 RVS uncertainties are intrinsically larger than $2\,\mathrm{km\,s^{-1}}$ for many of the cool giants that dominate the sample at the typical distance, where the magnitude-dependent inflation of σ_{RV} (Katz et al. 2023) interacts with the $\sim 25\,\mathrm{km\,s^{-1}}$ velocity scale of interest. Within the surviving subset the median eccentricity remains high (0.968 vs. 0.971 for the full Gold sample), the median $V_{\rm grf}$ is similar ($16.57\,\mathrm{km\,s^{-1}}$ vs. $15.72\,\mathrm{km\,s^{-1}}$), and 100% retain $e > 0.8$. The high-eccentricity claim survives this additional quality cut.

B. INNER-GALAXY CLOSE APPROACHES

The most extreme central-passage claims require additional caution. In the point-estimate products, the closest static and barred trajectories reach within 3.1 and 4.3 pc of the Galactic centre, respectively. Dedicated 5,000-realisation Monte Carlo propagation of the closest static and barred seeds shows that no candidate has $P(r_{\rm sph,min} < 10\,\mathrm{pc}) > 0.5$, while six candidates have $P(r_{\rm sph,min} < 100\,\mathrm{pc}) > 0.5$. These probabilities use the spherical Galactocentric closest approach, $r_{\rm sph} = \sqrt{x^2 + y^2 + z^2}$, and are Monte Carlo posterior probability counts, not the point-estimate cylindrical $R_{\rm peri}$ counts above. Since $r_{\rm sph} \geq \sqrt{x^2 + y^2}$, the Sgr A* set is a stricter subset of the in-plane central-reach set. We therefore describe these as candidate inner-Galaxy approachers, not as a confirmed population of Sgr A*-passing stars. A three-point quadratic interpolation of $r^2_{\rm sph}(t)$ around each sampled trajectory minimum is included in the probabilities in Table 30. Figure 21 shows the corresponding cumulative probability curves for the 20 closest point-estimate seeds in the static and barred-default integrations.

Table 27. Galactic-potential sensitivity for the catalogue. Values, including the eccentricity columns, use point-estimate integrations for all Tier A+B+C stars; all rows use the same 40,001-point, parabola-interpolated trajectory readout. $N_{\rm bridge}$ counts stars with $R_{\rm peri} < 2\,\mathrm{kpc}$ and $R_{\rm apo} > 15\,\mathrm{kpc}$. Barred median pericentres are resolution-limited readouts at this sampling; long-apocentre inner-reach and central-reach counts are the stable barred diagnostics. The McMillan+2017 row uses the independently calibrated, Gaia-DR3-external McMillan (2017) potential and reproduces the static Hunter+2024 result.

Variant	$\tilde{e}$	$e > 0.95$ (%)
Hunter+2024 static	0.964	66.7
McMillan+2017 static	0.961	62.8
Halo mass 0.85×	0.963	64.4
Halo mass 1.15×	0.965	68.3
Barred $\Omega_p = 24$	0.995	93.7
Barred $\Omega_p = 28$	0.996	93.0
Bar angle sweep	≈ 0.996	91.0–95.4
Halo flattening sweep	0.963–0.965	64.4–68.3
Portail+2017 M2M bar	0.996	92.7

Variant	$\widetilde{R_{\rm peri}}$ (pc)	$\widetilde{R_{\rm apo}}$ (kpc)	$N_{\rm br}$
Hunter+2024 static	116	7.36	3
McMillan+2017 static	123	7.31	3
Halo mass 0.85×	119	7.37	3
Halo mass 1.15×	111	7.33	3
Barred $\Omega_p = 24$	$\lesssim 14$	7.89	8
Barred $\Omega_p = 28$	$\lesssim 14$	7.92	7
Bar angle sweep	$\lesssim 13$–$\lesssim 15$	8.30–8.45	17–22
Halo flattening sweep	111–119	7.33–7.37	3–3
Portail+2017 M2M bar	$\lesssim 14$	8.38	15

C. ACTION-SPACE AND BAR RESONANCES

The same L_z axis is the natural place to read off where the catalogue's angular-momentum distribution sits relative to the bar resonances. Figure 22 plots the Tier A+B+C catalogue in L_z–J_R Stäckel-fudge action space and overlays the inner Lindblad, corotation, and outer Lindblad loci derived from the axisymmetrised Hunter+2024 potential for the full pattern-speed grid $\Omega_p \in \{24, 28, 33, 37.5, 41\}\,\mathrm{km\,s^{-1}\,kpc^{-1}}$. The Tier A points concentrate near $L_z \approx 0$ at large J_R, intersecting the ILR loci of several pattern speeds rather than piling up at any single corotation or OLR; this is the same low-angular-momentum, high-radial-action signature expected of compact-pericentre orbits, and

Table 28. Solar-parameter sensitivity of the Tier-A/B/C catalogue counts and point-estimate median orbital geometry. The listed variants span the literature-quoted range of R_0, $z_\odot$, V_c, and the LSR motion (GRAVITY Collaboration 2022; Schönrich 2012; Reid & Brunthaler 2020). Tier counts vary by $\sim$ 50% across the variants, but ensemble orbital geometry is stable to $\sim$ 12% in $R_{\rm peri}$ and $\sim$ 1% in $R_{\rm apo}$ and eccentricity. Tier counts in this table are computed on the legacy preselection subsample used for the solar-frame re-tiering, so the absolute counts differ from the full-catalogue primary count (Tier A/B/C increments 289/252/1411; Table 1). The adopted catalogue default is the 229 km s^{-1} convention documented in Section 2.4; the 232 km s^{-1} row is retained only as a legacy working-convention sensitivity row. The stable geometry medians are the quantity of interest here.

Variant	R_0 (kpc)	V_c (km s^{-1})	$N_{\rm A}$	$N_{\rm B}$	$N_{\rm C}$	Med. $R_{\rm peri}$ (pc)	Med. $R_{\rm apo}$ (kpc)
legacy 232	8.178	232.0	215	116	300	150	8.10
GRAVITY+22	8.275	229.0	234	132	355	154	8.14
LSR-6 (S12)	8.178	232.0	249	156	365	141	7.99
RB20	8.178	248.5	176	100	239	163	8.10

Table 29. Sensitivity battery status for the Tier A+B+C catalogue (N = 1,952). The default static and barred point-estimate orbit products have been regenerated from the parent-complete candidate pool. All median-e entries are point-estimate orbit summaries. Halo-mass and bar-pattern-speed rows are point-estimate sensitivity runs using the same initial conditions; the distance-estimator row remains a non-orbit tiering stress test.

Axis	Variant	Med. $R_{\rm peri}$ (pc)	Med. $R_{\rm apo}$ (kpc)	Med. e	$N_{\rm reach}$
Default static	Hunter+2024 axisymmetric	116	7.36	0.964	3
Default barred	Hunter/Sormani barred, $\Omega_p = 37.5$	$\lesssim 14$	8.32	0.996	21
Halo mass	$0.85 \times M_{\rm halo}$	119	7.37	0.963	3
Halo mass	$1.15 \times M_{\rm halo}$	111	7.33	0.965	3
Bar pattern speed	$\Omega_p = 33$ km s^{-1} kpc^{-1}	$\lesssim 15$	8.28	0.996	8
Bar pattern speed	$\Omega_p = 41$ km s^{-1} kpc^{-1}	$\lesssim 13$	7.97	0.996	31
Distance estimator	BJ/parallax coupling stress test	—	—	—	no tier-scale change

Note—The default rows use the higher-resolution orbit products reported in the main orbit table. Halo-mass and bar-speed rows use a uniform 40,001-point, parabola-interpolated sensitivity integration. The barred median pericentre readouts continue to fall between 20,001 and 40,001 points, so they are quoted as resolution-limited diagnostics; the long-apocentre inner-reach counts and $R_{\rm peri} < 100$ pc counts are the stable barred comparisons.

it is consistent with the matched-control OLR-fraction check in Section 4.2.

We stress that this overlay is qualitative. The Stäckel-fudge actions on which it is built carry a large vertical-action uncertainty for these highly eccentric, inner-plunging orbits—the median absolute fractional J_z difference between catalogue and orbit-time-averaged actions is 85% in Tier A and 62% over the full Tier A+B+C sample, against $\lesssim$ 10% for J_R and L_z (Appendix A)—so the figure should be read as locating the catalogue's low-L_z, high-J_R concentration relative to the resonance loci, not as a quantitative resonance-occupancy classification. The weighted matched-control comparison does not favour a simple outer-Lindblad pile-up explanation.

D. PRESENT-DAY CLUSTERING NULL TEST

As a final present-day compactness check, we searched the slow catalogue for open-cluster- or association-like aggregates in 3D Galactocentric position (Table 31). The Tier A+B sample contains one pair below 25 pc, with a closest-pair separation of 20.2 pc. The broader Tier A+B+C sample contains two close pairs below 25 pc, but a DBSCAN sensitivity grid (Ester et al. 1996; Schubert et al. 2017) over $\varepsilon = \{5, 10, 15, 25, 50\}$ pc and `min_samples`= $\{3, 4, 5, 7, 10\}$ returns zero compact groups for both Tier A+B and Tier A+B+C. Thus the catalogue is not explained by one or more present-day bound clusters or compact associations. This does not rule out common accretion-related origins or action-

Table 30. Candidate Sgr A* approachers from the Tier A+B+C catalogue. Candidate rows are seeded from the closest point-estimate static and barred integrations, then re-evaluated with 5,000-realisation Monte Carlo orbit propagation and three-point quadratic interpolation of $r^2_{\rm sph}(t)$ around each sampled minimum.

Gaia DR3 source_id	Tier	Potential	$r_{\rm sph,min}$ p16 (pc)	p50	p84	$P(<10\,{\rm pc})$ / $P(<100\,{\rm pc})$
132425700141447168	C	barred	26	43	72	0.023 / **0.976**
5871390894093751936	B	static	16	40	78	0.072 / **0.951**
4047819072313279616	C	barred	20	43	84	0.041 / **0.895**
132425700141447168	C	static	17	48	96	0.069 / **0.867**
4047405312345623296	C	barred	23	53	172	0.027 / **0.754**
6761283447501038848	C	barred	27	70	193	0.023 / **0.643**
4036336430101629568	C	static	18	110	337	0.079 / 0.475
4038063281786427520	C	static	31	117	312	0.037 / 0.463
4074796625207486080	C	static	83	147	361	0.008 / 0.252
6629652011339174912	C	barred	90	222	349	0.003 / 0.179

Note—No catalogue candidate satisfies $P(r_{\rm sph,min} < 10\,{\rm pc}) > 0.5$. Six candidates satisfy the softer $P(r_{\rm sph,min} < 100\,{\rm pc}) > 0.5$ criterion. These are spherical Galactocentric closest-approach probabilities from the Monte Carlo propagation and are distinct from the point-estimate cylindrical $R_{\rm peri} < 100\,{\rm pc}$ central-reach counts in Table 21. The closest point-estimate barred candidate is `6761283447501038848` with 4.3 pc. A source may appear twice when it is selected independently by the static and barred seed lists.

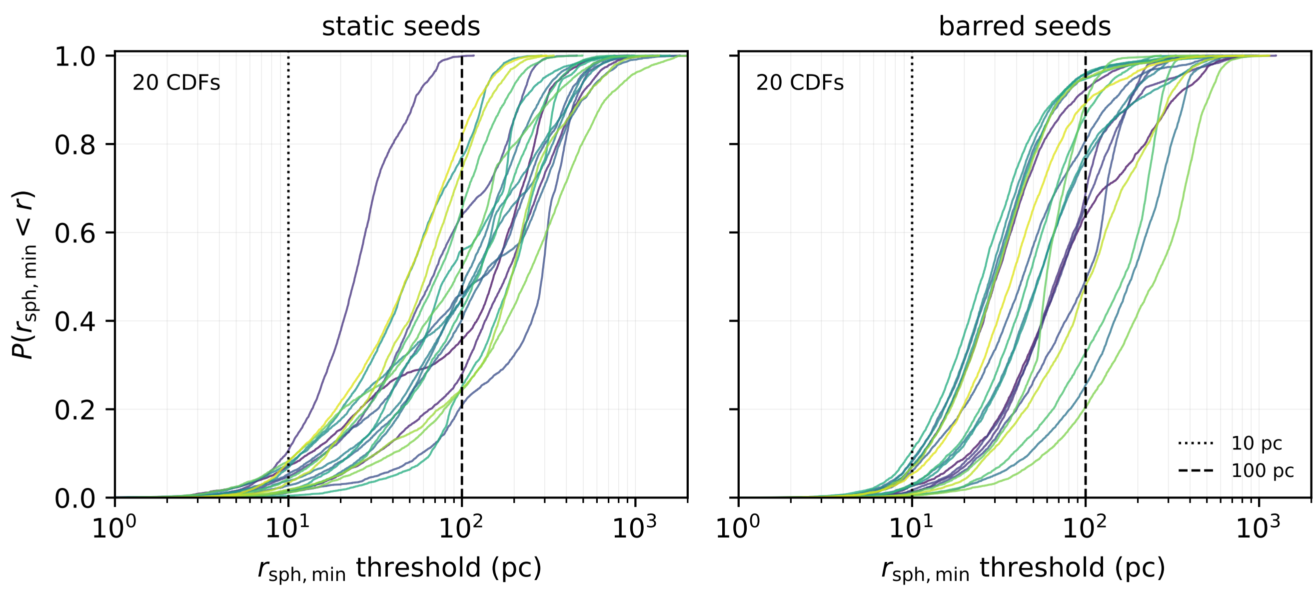

Figure 21. Monte Carlo cumulative probabilities for the 20 closest Sgr A* approachers in the static Hunter+2024 integration (left) and the barred-default integration (right). Each curve uses 5,000 propagated realisations and the same three-point interpolation of the sampled closest approach used in Table 30. The vertical reference lines mark 10 and 100 pc.

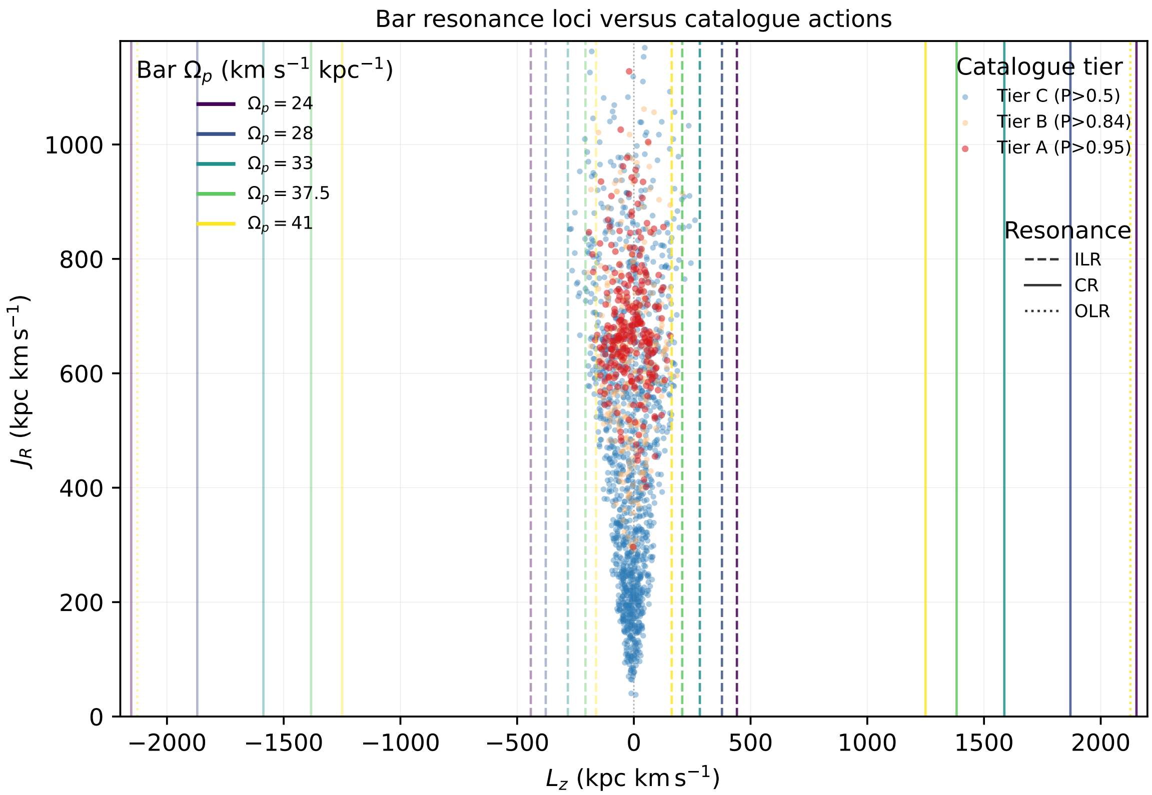

Figure 22. Bar resonance loci versus catalogue actions. Vertical lines mark the values of L_z for circular orbits at the inner Lindblad (dashed), corotation (solid), and outer Lindblad (dotted) radii in the axisymmetrised Hunter+2024 potential at each of the five bar pattern speeds in the extended grid (colour-coded). Loci at negative L_z mirror the retrograde-orbit case. Catalogue points are colour-coded by tier and use the Stäckel-fudge (L_z, J_R) values reported per star; Tier A points cluster near $L_z \approx 0$ at large J_R.

space substructure, which need not remain compact in present-day position.

Table 31. DBSCAN present-day compact-group sensitivity grid for the low-$V_{\rm grf}$ catalogue. Entries give the number of compact groups found in 3D Galactocentric position space for Tier A+B / Tier A+B+C.

ε (pc)	`min_samples`				
	3	4	5	7	10
5	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0
10	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0
15	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0
25	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0
50	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0

Note. The search uses the Ester et al. (1996) DBSCAN algorithm in present-day 3D Galactocentric position space, with ε varied from 5 to 50 pc and `min_samples` from 3 to 10. The paired entries are Tier A+B / Tier A+B+C group counts. The closest Tier A+B pair is 20.2 pc, while Tier A+B+C contains two pairs below 25 pc but no three-member group at any grid point. This grid follows the parameter-sensitivity guidance of Schubert et al. (2017).

E. DISTANCE-COVARIANCE TIER CONFUSION MATRIX

The tier-label confusion matrix under the maximal parallax–distance copula stress test of Section 2.3 is given below.

Table 32. Tier-label confusion matrix for the full tier-sensitive band ($N = 3,342$ sources with adopted $P \geq 0.3$) under a maximal parallax–distance Gaussian-copula coupling, relative to an independent-distance recomputation at the same realisations. Rows are the independent-draw tier, columns the coupled-draw tier; off-diagonal entries are tier switches. Only 114 of 3,342 sources switch. Relative to the adopted primary catalogue, 527 of 541 Tier A+B members remain above $P = 0.84$ under the coupled draw (median $|\Delta P| = 0.0090$, max 0.0550).

indep.\copula	A	B	C	<C
A	287	7	0	0
B	4	235	8	0
C	0	13	1355	40
<C	0	0	42	1351

F. EXTERNAL LOW-AZIMUTHAL-VELOCITY COMPARISON

Filion et al. (2025) provide the closest published Gaia-era comparison sample selected by low Galactocentric azimuthal velocity rather than by low total Galactic rest-frame speed. Resolving their supplementary APOGEE identifiers to Gaia DR3 source identifiers gives 69 unique Gaia sources, all satisfying the present Gaia-RVS parent-source contract. Under the present low-$V_{\rm grf}$ diagnostic tiering, however, all 69 are Tier X: three enter the 20,829-source Monte Carlo candidate pool, but none enter Tier A, Tier A+B, or Tier A+B+C. Across the full Filion set, the same Gaia RVS quality audit classifies 66 sources as `ok` and three as `poor`; three have RV-variability flags and two are matched in the Gaia DR3 non-single-star two-body-orbit table. The three candidate-pool overlaps themselves are RVS-quality `ok` and NSS-clean, although one has RUWE just above 1.4. Thus the comparison separates two definitions of "slow": Filion et al. select low azimuthal rotation with disc-like chemistry, whereas the present catalogue selects low total $V_{\rm grf}$ independent of chemistry and applies the same probability and quality gates to every source.

Table 33. Filion et al. (2025) stars entering the present Monte Carlo candidate pool.

Gaia DR3 source_id	APOGEE ID	$V_\phi^{\rm F25}$ ($\rm km\,s^{-1}$)	$V_{\rm grf}$ ($\rm km\,s^{-1}$)	$P_{<25}$	$\sigma_{\rm RV}$ ($\rm km\,s^{-1}$)	$N_{\rm tr}$	$p_{\chi^2,\rm RV}$	RUWE	RV class
3685921348377044480	2M13041724-0134399	-37.5	27.5	0.364	1.17	18	–	0.95	ok
1594059695819370624	2M14575158+5300140	15.2	40.0	0.090	1.14	26	0.640	1.17	ok
1288956935783694464	2M15022633+3248296	-24.8	47.6	0.000	0.19	33	0.924	1.41	ok

Note. $V_\phi^{\rm F25}$ is an approximate reproduction of the Filion et al. Galactocentric azimuthal-velocity convention using APOGEE DR17 line-of-sight velocities and Bailer-Jones et al. (2021) photogeometric distances. The remaining columns are from the present Gaia DR3 RVS low-$V_{\rm grf}$ release products. None of the three sources is matched in the Gaia DR3 non-single-star two-body-orbit table. All three sources are classified as Tier X and therefore are excluded from the Tier A+B+C science catalogue.

REFERENCES

Abdurro'uf, Accetta, K., Aerts, C., et al. 2022, ApJS, 259, 35

Andrae, R., Fouesneau, M., Sordo, R., et al. 2023, A&A, 674, A27

Astropy Collaboration, Robitaille, T. P., Tollerud, E. J., et al. 2013, A&A, 558, A33

Bailer-Jones, C. A. L., Rybizki, J., Fouesneau, M., Demleitner, M., & Andrae, R. 2021, AJ, 161, 147

Bailer-Jones, C. A. L. 2022, ApJL, 935, L9

Belokurov, V., Erkal, D., Evans, N. W., Koposov, S. E., & Sherwin, B. D. 2018, MNRAS, 478, 611

Belokurov, V., Vasiliev, E., Deason, A. J., et al. 2023, MNRAS, 518, 6200

Binney, J., & Merrifield, M. 1998, Galactic Astronomy (Princeton, NJ: Princeton Univ. Press)

Bland-Hawthorn, J., & Gerhard, O. 2016, ARA&A, 54, 529

Cantat-Gaudin, T., Fouesneau, M., Rix, H.-W., et al. 2023, A&A, 669, A55

Castro-Ginard, A., et al. 2023, A&A, 677, A37

Castro-Ginard, A., Jordi, C., Luri, X., et al. 2020, A&A, 635, A45

Carballo-Bello, J. A., Ramos, P., Corral-Santana, J. M., et al. 2025, A&A, 700, A172

Deason, A. J., Belokurov, V., Koposov, S. E., & Lancaster, L. 2018, ApJL, 862, L1

Deason, A. J., & Belokurov, V. 2024, New Astron. Rev., 99, 101706

Dillamore, A. M., Davies, E. Y., Belokurov, V., & Evans, N. W. 2023, MNRAS, 524, 3596

Dillamore, A. M., Belokurov, V., & Evans, N. W. 2024, MNRAS, 532, 4389

Eilers, A.-C., Hogg, D. W., Rix, H.-W., & Ness, M. K. 2019, ApJ, 871, 120

Ester, M., Kriegel, H.-P., Sander, J., & Xu, X. 1996, in Proceedings of the Second International Conference on Knowledge Discovery and Data Mining, 226

Gaia Collaboration, Vallenari, A., et al. 2023, A&A, 674, A1

GRAVITY Collaboration, Abuter, R., Amorim, A., et al. 2019, A&A, 625, L10

Helmi, A., Babusiaux, C., Koppelman, H. H., et al. 2018, Nature, 563, 85

Helmi, A. 2020, ARA&A, 58, 205

Huchra, J. P., & Geller, M. J. 1982, ApJ, 257, 423

Hunt, J. A. S., Bovy, J., & Carlberg, R. G. 2016, ApJL, 832, L25

Katz, D., et al. 2023, A&A, 674, A5

Lee, A., Lee, Y. S., Kim, Y. K., et al. 2023, ApJ, 945, 56

Marchetti, T., Evans, F. A., & Rossi, E. M. 2022, MNRAS, 515, 767

Naidu, R. P., Conroy, C., Bonaca, A., et al. 2020, ApJ, 901, 48

Queiroz, A. B. A., Anders, F., Chiappini, C., et al. 2023, A&A, 673, A155

Recio-Blanco, A., de Laverny, P., Palicio, P. A., et al. 2023, A&A, 674, A29

Rix, H.-W., Chandra, V., Andrae, R., et al. 2022, ApJ, 941, 45

Rybizki, J., Demleitner, M., Bailer-Jones, C. A. L., et al. 2020, PASP, 132, 074501

Scholz, F. W., & Stephens, M. A. 1987, Journal of the American Statistical Association, 82, 918

Schubert, E., Sander, J., Ester, M., Kriegel, H.-P., & Xu, X. 2017, ACM Transactions on Database Systems, 42, 19

Schönrich, R., Binney, J., & Dehnen, W. 2010, MNRAS, 403, 1829

Sharpe, K., Naidu, R. P., & Conroy, C. 2024, ApJ, 963, 162

Viswanathan, A., Starkenburg, E., Koppelman, H. H., et al. 2023, MNRAS, 521, 2087

Zhang, X., Green, G. M., & Rix, H.-W. 2023, MNRAS, 524, 1855

Zhang, L., Feng, F., Rui, Y., Xiao, G.-Y., & Wang, W. 2025, Research in Astronomy and Astrophysics, 25, 055010

Binney, J. 2012, MNRAS, 426, 1324

Buder, S., Sharma, S., Kos, J., et al. 2021, MNRAS, 506, 150

Hunter, G. H., Sormani, M. C., Benítez-Llambay, A., et al. 2024, A&A, 692, A216

Lindegren, L., Bastian, U., Biermann, M., et al. 2021, A&A, 649, A4

Sanders, J. L., & Binney, J. 2016, MNRAS, 457, 2107

Vasiliev, E. 2019, MNRAS, 482, 1525

Belokurov, V., Sanders, J. L., Fattahi, A., et al. 2020, MNRAS, 494, 3880

Belokurov, V., & Kravtsov, A. 2022, MNRAS, 514, 689

Cautun, M., Benítez-Llambay, A., Deason, A. J., et al. 2020, MNRAS, 494, 4291

GRAVITY Collaboration, Abuter, R., Aimar, N., et al. 2022, A&A, 657, L12

Reid, M. J., & Brunthaler, A. 2020, ApJ, 892, 39

Schönrich, R. 2012, MNRAS, 427, 274

Wang, W., Han, J., Cautun, M., Li, Z., & Ishigaki, M. N. 2020, Science China Phys. Mech. Astron., 63, 109801

Cameron, E. 2011, PASA, 28, 128

Hainmueller, J. 2012, Political Analysis, 20, 25

Clarke, J. P., Wegg, C., Gerhard, O., Smith, L. C., Lucas, P. W., & Wylie, S. M. 2019, MNRAS, 489, 3519

Everall, A., & Boubert, D. 2022, MNRAS, 509, 6205

Filion, C., Petersen, M. S., Horta, D., Daniel, K. J., Lucey, M., & Price-Whelan, A. M. 2025, ApJ, 989, 70

Horta, D., Petersen, M. S., & Peñarrubia, J. 2024, MNRAS, 538, 998

Luri, X., Brown, A. G. A., Sarro, L. M., et al. 2018, A&A, 616, A9

Portail, M., Gerhard, O., Wegg, C., & Ness, M. 2017, MNRAS, 465, 1621

Sormani, M. C., Magorrian, J., Nogueras-Lara, F., et al. 2022, MNRAS, 514, L1

Woudenberg, H. C., & Helmi, A. 2025, A&A, 700, A240

Wright, T. J., & Binney, J. 2026, MNRAS, in press, https://doi.org/10.1093/mnras/stag839

Zhang, H., Belokurov, V., Evans, N. W., Kane, S. G., & Sanders, J. L. 2024, MNRAS, 533, 3395

Babusiaux, C., et al. 2023, A&A, 674, A32

McMillan, P. J. 2017, MNRAS, 465, 76

Palicio, P. A., Recio-Blanco, A., Poggio, E., et al. 2023, A&A, 670, L7